Neural FOXP2: Measuring and Steering the Default Response Language of Multilingual Language Models

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Abstract

When a prompt carries no cue about the language of the answer, multilingual language models usually reply in English, even when the deployment calls for another language. We study this *default response language*. First, we introduce an early-decoding defaultness score that predicts the language of the finished response from the first three decoding steps, and we validate it with fifteen tests of predictive, convergent, discriminant and known-groups validity, calibration, reliability and robustness to token-set construction. Second, we propose Neural FOXP2, an inference-time edit that selects language features in frozen public sparse autoencoders, compresses them into a low-rank subspace, and applies a signed update within a selected band of layers, leaving the model's weights unchanged. Across four instruction-tuned models and six target languages (Spanish, French, Chinese, Russian, Hindi and Arabic), Neural FOXP2 raises the share of responses written entirely in the target language from 1.9% to 58.7%. At matched distributional shift, the strongest activation-steering baseline reaches 44.7%; LoRA fine-tuning reaches 60.2%, but it shifts the output distribution further and costs more utility. Neural FOXP2 changes utility by -0.28 points on average across six benchmarks and leaves refusal rates unchanged within their confidence intervals.

1 Introduction

Ask a multilingual language model a question without saying which language you want the answer in, and it will almost always answer in English (Marchisio et al., 2024; Shani and Basirat, 2025). That is a failure whenever the deployment fixes the output language in advance, as for a regional assistant, a localized product or a public service, even though the same model writes fluently in the required language. English-dominated training data and tokenizers that need many more tokens for some languages than for English (Petrov et al., 2023) are commonly cited as contributing causes.

We call the language a model chooses in this situation its *default response language*, and ask two questions about it. Can the default be read off early in decoding, in a way that reliably reflects the language of the finished response? And can it be changed at inference time, cheaply, selectively and without degrading what the model already does well?

There are good reasons to expect an early, compact decision. Intermediate representations of several model families pass through an English-aligned space before becoming language-specific (Wendler et al., 2024), and decoding analyses find that the response language is largely settled within the first tokens (Nie et al., 2025). Existing remedies act less directly. Continued pretraining and multilingual instruction tuning (Lu et al., 2024; Üstün et al., 2024; Shaham et al., 2024) improve multilingual ability but require retraining. Prompting requires control over every request. Activation steering (Rimsky et al., 2024; Turner et al., 2024) and language-neuron editing (Tang et al., 2024) can change the output language, but they have not been compared under a common protocol that also accounts for leakage into other languages, utility and safety.

This paper answers both questions and makes three contributions. We begin with measurement and define a defaultness score that compares the probability mass a model places on target-language and English tokens over its first three decoding steps (Section 3). Because a score is only useful if it tracks what users eventually read, we validate it with fifteen tests on 24 model–language pairs (Section 3.3). Building on this score, we then introduce Neural FOXP2 (Section 4), a gradient-free edit that selects language features in frozen public sparse-autoencoder (SAE) dictionaries, compresses them into a low-rank subspace, and adds a signed update

to the residual stream inside a window of layers chosen on development data. Finally, we compare Neural FOXP2 under one protocol with prompting, activation steering and LoRA fine-tuning on four models and six languages, measuring language, leakage, utility, intent respect, safety and human judgments, and we test whether the model itself relies on the steering subspace (Section 6). Neural FOXP2 moves the default nearly as far as fine-tuning does, at a lower distributional cost and utility loss, and it does so more effectively than any other inference-time method we test.

2 Related Work

Language choice and language confusion. Multilingual models show systematic preferences for some languages (Shani and Basirat, 2025), carry English-like patterns into other languages (Guo et al., 2025), and perform unevenly across languages (Hu et al., 2025). Marchisio et al. (2024) introduce the Language Confusion Benchmark (LCB), which scores whether a response is in the requested language; we use it as an external criterion. Reasoning traces also drift toward English (Saji et al., 2026; Zhao et al., 2025). Unlike LCB, our score is computed from the first decoding steps, cheaply enough to guide a search over interventions.

Language-specific components. Language identity has been localized to neurons (Tang et al., 2024; Kojima et al., 2024; Gurgurov et al., 2025; Rahmanisa et al., 2025; Tezuka and Inoue, 2025) and to low-rank subspaces (Xie et al., 2022), and alignment training changes these representations (Zhang et al., 2026). Whether such components support cross-lingual behavior remains debated (Mondal et al., 2025; Zhao et al., 2024). We therefore keep only language-selective candidates that pass a causal test.

Activation steering and sparse autoencoders. Adding or removing residual-stream directions changes behavior without training (Turner et al., 2024; Rimsky et al., 2024; Zou et al., 2023; Li et al., 2023; Arditi et al., 2024), including the output language (Sundar et al., 2025; Ravisankar et al., 2026). Sparse autoencoders decompose activations into sparse, often interpretable features (Elhage et al., 2022; Huben et al., 2024; Bricken et al., 2023; Templeton et al., 2024), and several studies identify and steer language features in SAE dictionaries (Deng et al., 2025; Chou et al., 2025; Andrylie et al., 2026; Zhong et al., 2026; Al Ghussin et al., 2026; Wong et al., 2026). Wu et al. (2025) find that simple difference-in-means steering and prompting often outperform SAE-based steering; we therefore include both as baselines and compare all methods at matched distributional shift. Neural FOXP2 differs from single-feature SAE steering by combining a sparse feature support with a low-rank subspace and a layer window chosen on development data.

3 Measuring the Default Response Language

3.1 Setting

A deployment specifies a target language ℓ_t, and requests arrive without an instruction to use it. We evaluate three sets of held-out prompts. NO-CUE prompts are English requests that do not mention any language, drawn from seven task types: question answering, explanation, summarization, writing, advice, coding and reasoning. They are the everyday case and our main setting. NEUTRAL prompts contain no natural language at all (arithmetic, code, lists of names) and test whether the default persists without English in the input. INTENT prompts explicitly ask for English or are written in another suite language, and measure how far an edit overrides a clear preference. Sources, sizes and examples are in Appendix A.

3.2 Defaultness score

The idea behind the score is simple: if the model is about to answer in the target language, it should place probability on target-language tokens from the very first steps of decoding. Let V_{tgt} and V_{en} be the tokens diagnostic of the target language and of English for a given tokenizer, and let $w(u) \in [0, 1]$ down-weight tokens diagnostic of neither, such as digits, punctuation, shared subwords and named-entity fragments (Appendix E). For prompt x, decoding step t and class $c \in \{\mathrm{tgt}, \mathrm{en}\}$, the class mass is

$$M_c(x, t) = \sum_{u \in V_c} w(u)\, p_\theta\big(u \mid x, y_{<t}\big), \qquad (1)$$

where $y_{<t}$ is the greedy prefix. The defaultness score averages the difference between the two masses over a short horizon T and lies in $[-1, 1]$:

$$\Delta M(x) = \frac{1}{|T|} \sum_{t \in T} \big[M_{\mathrm{tgt}}(x, t) - M_{\mathrm{en}}(x, t)\big]. \qquad (2)$$

Figure placeholder

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Vector diagram, left to right. (1) Measure: a weak prompt, the next-token distributions at steps t=1..3, target vs English token mass, giving ΔM. (2) Stage I: SAE features at each layer scored by selectivity $\times$ causal lift; keep top-K per layer. (3) Stage II: SVD of target-minus-English feature differences gives a rank-r subspace per layer; choose window $\mathcal{W}$. (4) Stage III: signed edit (promote target, remove English component) decoded with W_{dec} and added to the residual stream inside $\mathcal{W}$. Show one example prompt and its output before and after the edit, with an English gloss.

Figure 1: **Overview.** The default response language is measured with an early-decoding score (left) and changed with a three-stage, gradient-free edit built from frozen public sparse autoencoders (right).

A positive score means that the model places more early probability on the target language than on English. We also consider a log-odds form and the target share $S(x) = \frac{1}{|T|}\sum_{t\in T} M_{\text{tgt}}/(M_{\text{tgt}}+M_{\text{en}}+\epsilon)$ (Appendix F.1). The form and the horizon were fixed on $\mathcal{D}_{\text{dev}}$ before any evaluation run, by selecting the form with the highest development AUC for the language of the finished response. That form is S (.848 against .843 for ΔM at three steps), which also agrees best with human language judgments (.827 against .814). We therefore use S as the primary predictive metric and keep ΔM as the mechanistically interpretable one: a signed difference of probability masses, linear in the next-token distribution, which serves as the objective of the causal probes in Stage I and as the basis of DEFAULT. We report both. For both forms the selected horizon is $T = \{1, 2, 3\}$.

Why three decoding steps. The horizon balances two risks. Too short, and the score sees only the first token, which can be a formatting mark or the start of a named entity and says little about the language that follows (Appendix F.10). Too long, and the score measures the consequences of a choice rather than the choice: later distributions are conditioned on a prefix already in one language, while every extra step costs a forward pass, also in each causal probe of Stage I. Three observations place the balance at three steps (Figure 2). First, prediction saturates: on $\mathcal{D}_{\text{dev}}$ the AUC of ΔM rises from .778 at one step to .817 at two and .843 at three, then stays flat (.843 at five, .851 at ten), so our pre-registered rule, the smallest horizon within .01 of the best AUC, selects three steps for both ΔM and S; language identification on the same tokens lags (.816) and needs ten to catch up. Second,

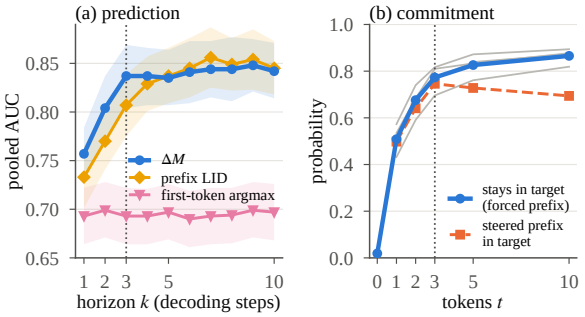


Figure 2: **Why three decoding steps.** (a) Pooled prompt-level AUC for predicting the language of the finished response from the first k steps ($\mathcal{D}_{\text{eval}}$, 95% bands; a different aggregation from the macro AUC of Table 13): ΔM saturates at k=3, whereas language identification on the prefix needs more tokens and the first-token argmax never catches up. (b) Blue: probability that an unedited response stays in the target language after its first t tokens are forced into it, mean over models; at t=0 it equals the unedited full-response rate (1.9%). Grey: per-model curves for $t \geq 1$. Orange: share of Neural FOXP2 responses whose first t tokens are in the target language. Commitment is largely in place, and the steered opening is strongest, at three tokens. Dotted line: the selected horizon.

the model commits within this window: when the first t tokens are forced into the target language, the share of responses that stay there rises from 1.9% with no forced target-language tokens to 77.3% after forcing three tokens, and only to 86.6% after ten, so three tokens deliver 89% of the increase reached with ten. Third, under Neural FOXP2 the share of target-language openings peaks at exactly three tokens (74.6%) and then declines; a longer horizon would blend the decision with later drift, which we measure separately through the full-response rate. Appendix F.2 gives the full analysis.

Outcome measures. Because ΔM is a proxy, we also report the language of the generated text. The *full-response rate*, our primary outcome, is the share of responses in which every sentence of at least 20 characters is identified as the target language. DEFAULT combines the score with the opening of the response: a prompt counts as a success when $\Delta M > 0$ and a language identifier (GlotLID; Kargaran et al., 2023) assigns the target language to the first **[20]** generated tokens. We also report the line-level pass rate of LCB (Marchisio et al., 2024). We evaluate the score against external outcomes that are not defined from ΔM, including full-response language labels, human judgments and task performance; DEFAULT itself depends on ΔM and is not treated as an independent validation outcome.

3.3 Validating the score

We validate ΔM with fifteen tests on all 24 model–language pairs (Appendix F), covering prediction and calibration, sensitivity to cue strength, reliability, discriminant and close-language validity, robustness to token sets, decoding and template, and an analysis of disagreements with the finished response.

Prediction and calibration. Three steps of probability mass largely stand in for a full response. At the selected horizon, ΔM predicts the language of the finished response with a macro AUC of .819 over the 24 pairs (Table 13 in Appendix F), and S agrees best with human judgments (.827). It outperforms every predictor of equal or lower cost, namely language identification on the first three tokens (.787), first-token argmax (.733) and the perplexity ratio (.724), and it is matched only by predictors that read far more output: identification on the first 20 tokens (.823) and an LLM judge of the full response (.834). The same ordering holds against human language judgments. Per-model isotonic maps calibrate ΔM to an expected calibration error of .048–.071, against .076–.110 for maps transferred across models (Table 16).

Construct validity. ΔM rises with the strength of a language cue, from -0.571 when the prompt has no cue to 0.030 when it is written entirely in the target language, monotonically for five of the six languages (**[Jonckheere–Terpstra statistic and p]**). It is reliable across paraphrases and token-set halves (ICC 0.71–0.82, split-half 0.85–0.92), nearly unrelated to prompt length, task type and answer quality (Table 17), and it separates close languages (Spanish ΔM is $+0.08$ on French responses, $+0.42$ on Spanish ones). Finally, our main result does not depend on how token sets are built: under Neural FOXP2, DEFAULT ranges from 0.495 to 0.583 across four constructions, while random token sets of matched size and frequency give DEFAULT values whose 95th percentile is at most 0.059.

3.4 How unedited models choose a language

Without an edit, every model answers in English. On NO-CUE prompts, mean ΔM is negative for all 24 pairs (macro -0.542; Table 20), and only 1.9% of responses are written entirely in the target language (1.6–2.4% across languages; Table 19). The early preference for English is strongest for Gemma-2-2B-it (mean ΔM -0.597) and weakest for Qwen3-8B (-0.458). Removing English from the input makes ΔM less negative without changing its sign (macro -0.453 on NEUTRAL prompts), and the full-response target rate stays at 1.6%. English is not merely echoed from the prompt; it is the fallback.

To write once results are in: A1: the NO-CUE rows of Table 20(b) and the NO-CUE cell rates of Figure 17 came from a run that gives a 1.55% macro and must be regenerated from Experiment 1 so that they average to 1.9% (es 1.6, fr 1.9, zh 2.2, ru 1.8, hi 2.4, ar 1.7). The NEUTRAL rows of the same table average 4.3%, not the canonical 1.6%; reconcile them too.

The choice is made early and persists: forcing three target-language tokens raises the share of responses that stay in the target language from 1.9% to 77.3%, and with three English tokens forced, 97.3% stay in English (Appendix G.1). The opening tokens strongly influence the language but do not determine it irreversibly: about 22.7% of continuations still leave the target language after a three-token forced opening. This commitment lets an early score predict the finished response, and it suggests where to intervene: an edit that shifts the first steps should carry through the response.

4 Neural FOXP2

Neural FOXP2 turns this observation into an intervention. It has three stages that use no gradients and update no weights. Stage I finds SAE features that are both language-selective and causally able to raise defaultness. Stage II compresses them into a low-rank subspace per layer and selects a contiguous window of layers in which to act. Stage III applies a signed edit inside that window. The result is a fixed forward hook per model and target language.

SAE dictionaries. We use the frozen, publicly released residual-stream SAEs of Llama Scope (He et al., 2024), Gemma Scope (Lieberum et al., 2024) and Qwen-Scope (Deng et al., 2026). All three suites were trained on activations of the base models and are applied here to the instruction-tuned models; Appendix B reports their reconstruction quality on instruction-tuned activations and compares them with the instruction-tuned SAEs that Gemma Scope provides for Gemma-2-9B-it. At layer ℓ, the encoder gives features $z^{(\ell)}(x) = E^{(\ell)}(h^{(\ell)}(x)) \in \mathbb{R}^m$ for the residual activation $h^{(\ell)}(x) \in \mathbb{R}^d$, and $W_{\text{dec}}^{(\ell)} \in \mathbb{R}^{d\times m}$ is the decoder matrix with columns d_j.

4.1 Stage I: language features

Discovery uses N meaning-matched pairs $(x_{\text{en}}^{(k)}, x_{\ell_t}^{(k)})$ from $\mathcal{D}_{\text{disc}}$ and activations at the final prompt token. The *selectivity* of feature j is a standardized mean difference,

$$\text{Sel}_j^{(\ell)} = \frac{\mathbb{E}_k\left[z_j^{(\ell)}(x_{\ell_t}^{(k)}) - z_j^{(\ell)}(x_{\text{en}}^{(k)})\right]}{\sigma_j^{(\ell,\ell_t)} + \sigma_j^{(\ell,\text{en})} + \epsilon}, \quad (3)$$

where the σ terms are within-language standard deviations. Because selectivity is correlational, we also measure *causal lift*, the change in ΔM per unit of α when αd_j is added to $h^{(\ell)}$ on weak prompts:

$$\text{Lift}_j^{(\ell)} = \underset{\alpha\in\{\alpha_1,\alpha_2,\alpha_3\}}{\text{median}} \frac{1}{\alpha}\mathbb{E}_{x\in\mathcal{D}_{\text{disc}}}\left[\Delta M_\alpha(x) - \Delta M(x)\right]. \quad (4)$$

Features are ranked by $\max(\text{Sel}_j^{(\ell)}, 0) \cdot \max(\text{Lift}_j^{(\ell)}, 0)$, so a feature must pass both criteria, and the top K per layer form the support $\mathcal{N}_{\ell_t}^{(\ell)}$.

4.2 Stage II: subspace and window

For each layer we stack the feature differences restricted to the support, $\Delta z_k^{(\ell)} = \Pi_{\mathcal{N}^{(\ell)}}\left(z^{(\ell)}(x_{\ell_t}^{(k)}) - z^{(\ell)}(x_{\text{en}}^{(k)})\right)$, into a matrix $\Delta Z^{(\ell)} \in \mathbb{R}^{N\times m}$ and compute its singular value decomposition $\Delta Z^{(\ell)} = U\Sigma V^\top$. The steering subspace $\mathcal{S}_{\ell_t}^{(\ell)}$ is spanned by the first r_ℓ right singular vectors, where $r_\ell = \min(\lceil r_{\text{eff}}\rceil, i^\star)$, the smaller of the ceiling of the effective rank and the index of the largest eigengap (Appendix C).

The intervention window $\mathcal{W}$ is the contiguous range of layers, among those whose subspace is stable under bootstrap resampling, that maximizes a development objective rewarding the defaultness gain and penalizing leakage, KL drift and utility loss (Appendix C).

4.3 Stage III: signed edit

Let $P^{(\ell)}$ project onto $\mathcal{S}_{\ell_t}^{(\ell)}$, let $\mu^{(\ell)} = \mathbb{E}_k[\Delta z_k^{(\ell)}]$ be the mean English-to-target shift, and let $\bar{z}^{(\ell)} = \mathbb{E}_{x\in\mathcal{D}_{\text{disc}}}[\Pi_{\mathcal{N}^{(\ell)}} z^{(\ell)}(x)]$ be the mean feature state on English prompts. With the alignment coefficient

$$a^{(\ell)}(x) = \frac{\langle \Pi_{\mathcal{N}^{(\ell)}} z^{(\ell)}(x),\ \bar{z}^{(\ell)}\rangle}{\|\bar{z}^{(\ell)}\|_2^2 + \epsilon}, \quad (5)$$

the edit for $\ell \in \mathcal{W}$ is

$$\delta z^{(\ell)}(x) = \Pi_{\mathcal{N}^{(\ell)}}\left(\lambda P^{(\ell)}\mu^{(\ell)} - \beta\, a^{(\ell)}(x)\,\bar{z}^{(\ell)}\right), \quad (6)$$

and the residual stream is updated additively through the decoder,

$$h^{(\ell)}(x) \leftarrow h^{(\ell)}(x) + \gamma\, W_{\text{dec}}^{(\ell)}\,\delta z^{(\ell)}(x). \quad (7)$$

The first term promotes the target language along the stable subspace; the second removes the part of the current state aligned with the English mean. Because the update adds a vector and omits the decoder bias, SAE reconstruction error never enters the forward pass, and $\gamma = 0$ reproduces the unedited model exactly. At the operating point, $\beta/\lambda = 1$ for every pair and γ is chosen per model–language pair on $\mathcal{D}_{\text{dev}}$ from $\{0.25, 0.5, 0.75, 1, 1.5, 2, 3\}$ under the guardrails (median 0.75). All hyperparameters are listed in Table 8.

5 Experimental Setup

Models and languages. We steer Llama-3.1-8B-Instruct (Grattafiori et al., 2024), Gemma-2-9B-it and Gemma-2-2B-it (Gemma Team, 2024), and Qwen3-8B (Yang et al., 2025) with thinking mode disabled; their families have public per-layer residual SAE suites. The target languages, Spanish, French, Chinese, Russian (high-resource), Hindi and Arabic (medium-resource), cover five scripts and both writing directions; every experiment is run on all 24 model–language pairs.

Data. Discovery and development pairs ($\mathcal{D}_{\text{disc}}$, $\mathcal{D}_{\text{dev}}$) come from the FLORES-200 `dev` split and evaluation pairs ($\mathcal{D}_{\text{eval}}$) from `devtest` (Team et al., 2022). Pairs pass an entailment filter audited by native speakers (Table 2). The NO-CUE, NEUTRAL and INTENT sets contain 500, 200 and **[n]** evaluation prompts per language.

Methods compared. We compare Neural FOXP2 with eight baselines, alongside the unedited model as a control and the combination of Neural FOXP2 with the system prompt, all run with the same prompts, decoding settings and evaluation harness (Appendix D).

Prompting. The *short instruction* appends "Answer in *X*." to the user's request. The *system prompt* places a single fixed instruction, "Always respond in *X*.", in the system turn. It is never tuned. The *best prompt* is a tuned upper bound: for every model–language pair we keep the best of ten strategies on $\mathcal{D}_{\text{dev}}$, including the system prompt, exemplars, personas and instructions in the target language (Table 10).

Activation steering. Difference-in-means adds the mean target-minus-English residual difference in the Neural FOXP2 window (Rimsky et al., 2024), its scale tuned per pair on $\mathcal{D}_{\text{dev}}$ (median γ 0.75); *with projection* it also removes the English direction (Arditi et al., 2024), a dense analogue of Eq. 6; *LAPE neurons* steers language-specific FFN neurons (Tang et al., 2024); *SAE features* adds the decoder directions of the most selective features, without lift, subspace or window (Chou et al., 2025).

Weight update. LoRA fine-tunes adapters (Hu et al., 2022) on the model's own target-language answers to discovery prompts.

Steering methods are tuned on $\mathcal{D}_{\text{dev}}$ under the same guardrails and budget as Neural FOXP2 (Table 11).

Metrics. Besides the measures of Section 3.2, we report *leakage*, the mean absolute increase in off-target DEFAULT, expressed as a fraction, over the five non-target suite languages under an edit; *KL drift*, the mean KL divergence between edited and unedited next-token distributions over the first three steps; and the utility change ΔS, the predeclared mean of the changes on six benchmarks, four with target-language input and two with English input while the edit is active (Section 6.5). Definitions are in Appendix E.

Statistics. Decoding is greedy. We report 95% bootstrap intervals over prompts ($B = 2{,}000$) for the prespecified primary outcomes and main comparisons; secondary descriptive tables omit intervals where noted. Headline comparisons use paired permutation tests with Holm correction (Holm, 1979), with DEFAULT and the full-response rate as separate families (Table 48).

6 Results

6.1 Main comparison

Three patterns stand out in Table 1. First, prompting helps far less than expected: the best prompt reaches 26.3% and the system prompt 24.5%. Second, steering does considerably better, and Neural FOXP2 does best among inference-time methods. It raises the full-response rate from 1.9% to 58.7%, more than twice the rate of either prompt and 12.7 points above the strongest dense steering method, difference-in-means with projection (46.0%). Third, only fine-tuning comes close: LoRA reaches 60.2%, with overlapping intervals, but with 1.7 times the KL drift, almost four times the utility loss, a lower LCB pass rate and the need for training. On NEUTRAL prompts Neural FOXP2 is ahead (61.1% vs 58.4%; Table 25). Adding the system prompt to Neural FOXP2 raises the rate to 65.0%, against 24.5% for the system prompt alone. **[Holm-adjusted p for Neural FOXP2 against the best prompt, difference-in-means with projection and LoRA.]**

Sweeping each method's strength (Figure 3) confirms this. Neural FOXP2 reaches a higher full-response rate than every other steering method at every level of KL drift and every level of utility loss, except at the smallest strengths, where all methods remain below 7%. Interpolated per pair at the KL drift of the Neural FOXP2 operating point, the other steering methods reach 39.3–44.7% (Table 26), and none exceeds 54% at any strength, whereas Neural FOXP2 reaches 65%.

6.2 Languages and selectivity

Neural FOXP2 improves every language and every one of the 24 model–language pairs. Averaged over models, its full-response rate ranges from 66% for Spanish to 50% for Arabic (Table 19). Every method that changes the default ranks the six languages in the same order, from Spanish, French and Russian to Chinese, Hindi and Arabic, so the differences reflect the languages rather than the method. Across pairs the gain ranges from 31.8 to 69.5 points; even the smallest (Gemma-2-2B-it, Arabic) has a 95% CI of [23.1, 40.5].

The edits are also selective (Figure 4). On NO-CUE prompts each edit raises DEFAULT in its own language by 33.1 to 42.8 points (mean 38.2), mostly at the expense of English, while no non-target suite language rises by more than 2.1 points; averaged over the off-diagonal cells, leakage is 0.65

Method	Mean $\Delta M\uparrow$ (macro over languages)				Macro over 24 model–language pairs					
	Llama 8B	Gemma 9B	Gemma 2B	Qwen 8B	Def.↑	Full %↑	LCB↑	Leak.↓	KL↓	$\Delta S\uparrow$
No edit	−0.54	−0.57	−0.60	−0.46	[?]	$1.9_{\pm 0.7}$	0.61	0.000	0.000	0.0
Prompting										
Short instruction	−0.34	−0.40	−0.44	−0.29	[?]	$15.7_{\pm 1.9}$	0.65	[?]	0.011	−0.1
System prompt	−0.22	−0.27	−0.31	−0.16	[?]	$24.5_{\pm 2.2}$	0.68	[?]	0.024	−0.3
Best prompt	−0.18	−0.24	−0.29	−0.13	[?]	$26.3_{\pm 2.4}$	0.69	[?]	0.027	−0.2
Activation steering										
Difference-in-means	−0.10	−0.14	−0.19	−0.05	[?]	$40.0_{\pm 2.6}$	0.73	[?]	0.079	−0.4
Diff.-in-means + proj.	−0.04	−0.09	−0.13	0.00	[?]	$46.0_{\pm 2.8}$	0.75	[?]	0.086	−0.5
LAPE neurons	−0.11	−0.16	−0.20	−0.06	[?]	$37.5_{\pm 2.5}$	0.72	[?]	0.069	−0.2
SAE features	−0.07	−0.11	−0.16	−0.02	[?]	$42.2_{\pm 2.7}$	0.74	[?]	0.091	−0.7
Weight update										
LoRA fine-tuning	0.02	−0.03	−0.09	0.06	[?]	$60.2_{\pm 3.0}$	0.78	[?]	0.125	−1.1
Neural FOXP2	0.01	0.04	−0.02	0.12	0.587	$58.7_{\pm 2.9}$	0.82	0.0065	0.073	−0.3
Neural FOXP2 + system prompt	0.07	0.06	0.03	0.15	[?]	$65.0_{\pm 2.8}$	0.83	[?]	0.082	−0.4

Table 1: **Main comparison** on English prompts without a language cue (NO-CUE, $\mathcal{D}_{\text{eval}}$), each method at its operating point. Def.: DEFAULT (fraction); Full %: full-response rate, mean$_{\pm 95\%\text{ CI half-width}}$; Leak.: mean absolute increase in off-target DEFAULT (fraction); other columns report means. Red cells are pending recomputation from the evaluation outputs. KL in nats, ΔS in points. Results on NEUTRAL prompts are in Table 25, results by language in Table 19.

points (0.0065). On NEUTRAL prompts, a different prompt set and not an estimate of the same quantity, the per-language gains are 48.3 to 64.4 points (mean 56.7).

To write once results are in: A3: the 48.3–64.4 range is computed from Table 19, which is captioned as NO-CUE and whose averages match Table 1 (protocol identity). Confirm that the range comes from a NEUTRAL run and cite that table, or remove the sentence.

Even Spanish and French, which share a script, move by only 1.8 and 2.1 points under each other's edit. Nor are they script edits: native-script Hindi and Russian edits also raise romanized early mass (3.3 and 3.6 points; Appendix H.1).

6.3 What each component contributes

Removing one element of Neural FOXP2 at a time lowers DEFAULT in every case (Table 27 in Appendix H). Suppression carries more of the effect: the suppression-only edit (λ=0) loses 6.5 points, the promotion-only edit (β=0) 24.6, consistent with the sensitivity sweep, where β/λ=0 gives a DEFAULT of 0.348 (Appendix I). Both stages of the construction matter. Dropping the subspace or the Stage I support costs 5.6 and 9.5 points and raises KL drift, consistent with their role of filtering out noisy or non-specific directions. Where the edit is applied matters as much as what it contains: shifting the window earlier or later costs 11.0–13.6 points. Random supports, subspaces and random edits matched in entropy change or norm lose 22.0–36.3 points, so the effect is not a generic perturbation. **[95% CIs of each drop.]** The language shift itself is low-rank: at the central window layer its effective rank is 8.4, against 12.8 for random supports and 18.4 for meaning-shuffled pairs, and for 20 of the 24 pairs it falls below the 5th percentile of the shuffled-pair null (Table 7).

6.4 Is the subspace used for language choice?

Steering shows that $\mathcal{S}$ is *sufficient* to change the output language, not that the model *uses* it. To test use, we reverse the experiment: we project $\mathcal{S}_{\ell_t}^{(\ell)}$ out of the unedited model inside $\mathcal{W}$ and ask it to follow an explicit instruction, "Answer in *X*." (Table 31). Removing it makes the instruction harder to follow: compliance falls from 15.7%, the full-response rate of the short instruction in Table 1, to 8.3%, whereas it stays at 13.1% with a random subspace of the same rank and at 12.4% when $\mathcal{S}$ is removed outside the window. The projection costs only 0.5–1.6 points on XNLI and MMLU (Table 33). **[Paired differences against the random-subspace and outside-window controls, with 95% CIs and Holm-adjusted p.]** The subspace is thus one route by which the model follows a language instruction, though not the only one.

6.5 Utility, intent and safety

A language change is of little value if answers get worse, so we measure utility on target-language inputs, on English inputs with the edit active, and in the NO-CUE answers themselves (Table 39). Neural FOXP2 leaves target-language tasks essentially

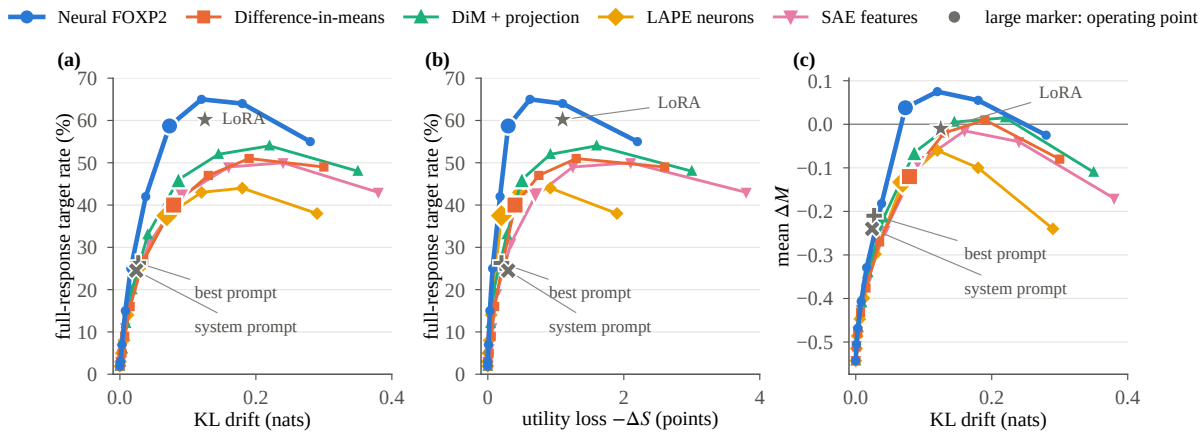


Figure 3: **Defaultness against cost.** Each curve sweeps one steering method's strength, macro over 24 model–language pairs; large markers are the operating points of Table 1, and the best prompt, the system prompt and LoRA are single points. (a) Full-response rate against KL drift. (b) Full-response rate against utility loss. (c) Mean ΔM against KL drift. Curves further up and to the left reach a higher rate at the same cost.

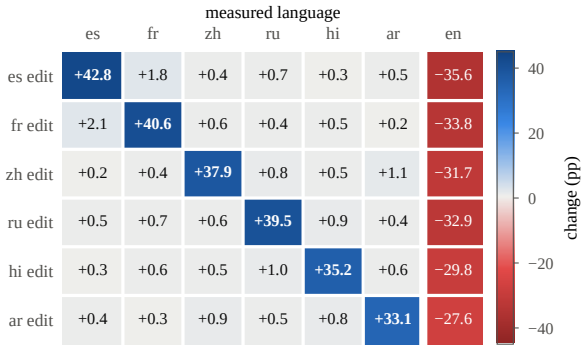


Figure 4: **Edits are language-selective.** Change in DEFAULT (percentage points) against the unedited model for every suite language under each language's Neural FOXP2 edit (NO-CUE, $\mathcal{D}_{\text{eval}}$); off-diagonal mean 0.65 points, i.e. leakage 0.0065, macro over four models; the en column is the change in fully English responses.

unchanged (mean -0.08 points, 95% CI $[-0.25, +0.10]$) and costs 0.57 points on English inputs ([95% CI]), for an overall change of -0.28 points ($[-0.42, -0.14]$) across the six benchmarks. Only 4 of the 144 task–pair combinations exceed the tolerance fixed on $\mathcal{D}_{\text{dev}}$. LoRA, by contrast, loses 0.8–1.4 points on every benchmark.

Refusal rates do not change measurably (Table 40): -0.8 points on target-language XSafety prompts ($[-2.7, +1.1]$), -0.5 on English harmful prompts and $+0.9$ for over-refusal on XSTest, with both intervals including zero. LoRA, by contrast, lowers refusal by about 5 points and raises over-refusal by 4.8.

Like a system prompt, Neural FOXP2 overrides explicit requests for English (63.2% of such INTENT prompts, against 75.4% for the system prompt) while keeping natively written prompts in their own language (82.9%; Table 43), so it suits deployments in which the response language is fixed.

6.6 Human evaluation

Native speakers rated the outputs of Neural FOXP2, the best prompt and difference-in-means steering without knowing which method produced them; each item received two or three independent ratings from native speakers of the target language (Table 45). They judged 60.7% of Neural FOXP2 responses fully in the target language, against 32.4% for the best prompt and 42.8% for difference-in-means (Holm $p = .008$, $.014$), with fluency and adequacy within 0.2 points of both (not significant) but lower naturalness than the best prompt (3.63 vs 4.02, Holm $p = .018$). Annotators chose the Neural FOXP2 output as the best most often (43%; Appendix K).

7 Conclusion

Multilingual models fall back to English when nothing in a request says otherwise. This choice can be read from the first three decoding steps, where prediction saturates, models commit and a steered opening is strongest. Building on this early decision, Neural FOXP2 raises the full-response rate from 1.9% to 58.7% across four models and six languages, against 44.7% for the strongest steering baseline at matched shift and 60.2% for LoRA at a higher shift, with a utility change of -0.28 points and no measurable change in refusals.

Limitations

The score is a proxy. ΔM is computed from the first decoding steps and inherits the imperfections of token sets under shared subwords, transliteration and code-mixing. The main behavioral comparisons therefore include full-response language and LCB outcomes, with human judgments used for the specified validation analyses; ablation, sensitivity, transfer and robustness experiments primarily report DEFAULT unless otherwise stated. We also analyze the cases in which ΔM and the finished response disagree (Appendix F.10). Writing in a language is also not writing it well, so fluency, adequacy and naturalness are reported separately.

Coverage. All six languages are high- or medium-resource, and the largest model has 9B parameters. Behavior on low-resource languages, unseen scripts and larger models is untested. The method requires a public SAE suite for the model family, so it cannot be applied to models without one, including closed models.

Dictionaries trained on base models. Llama Scope, Gemma Scope and Qwen-Scope were trained on base-model activations and are applied here to instruction-tuned models. The additive update keeps the forward pass exact when the edit is zero, but feature quality may be lower than on the base models (FVU 0.27–0.39 in the window; Table 5). For Gemma-2-9B-it, where Gemma Scope also provides instruction-tuned dictionaries, swapping them in changes DEFAULT, the full-response rate and utility by amounts whose 95% intervals include zero (Table 6).

Scope of claims. Our claims concern the default language under the prompt conditions we test. We do not claim that Neural FOXP2 adds linguistic competence, separates language from meaning, or that the selected features form a single disentangled language factor. Like a system prompt, Neural FOXP2 overrides explicit requests for English (63.2% of such INTENT prompts, against 75.4% for the system prompt), so it should not be left active where users may legitimately ask for English.

Ethical Considerations

Controlling the response language can improve access for speakers of languages that current defaults under-serve. The same control is dual-use: safety behavior is uneven across languages, so steering into a language with weaker safeguards is a plausible misuse, as is suppressing a language at scale. We measure refusal and over-refusal with and without the edit (Table 40), but these checks do not certify safety under adversarial use, and we recommend in-language safety evaluation before deployment. Human annotators were native speakers recruited through **[platform]**, paid **[rate]** per hour, above the local minimum wage, and gave informed consent; no personal data were collected (Appendix K). All datasets are used under their licenses (Table 2).

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A Data and Prompt Sets

Meaning-matched pairs. Discovery and evaluation pairs come from FLORES-200 (Team et al., 2022). $\mathcal{D}_{\text{disc}}$ and $\mathcal{D}_{\text{dev}}$ are drawn from the `dev` split and $\mathcal{D}_{\text{eval}}$ from `devtest`. Sentences used for the FLORES translation task in the utility evaluation are excluded from all three. A pair is kept only if a multilingual entailment model judges each side to entail the other with probability at least $\tau_{\text{sem}} = $ **[0.85]**. For each language, a native speaker independently judged 100 random pairs as keep or drop; Cohen's κ (Cohen, 1960) between this judgment and the filter is reported in Table 2.

Lang.	$\mathcal{D}_{\text{disc}}$	$\mathcal{D}_{\text{dev}}$	$\mathcal{D}_{\text{eval}}$	**QC%**	κ	**Script**
es						Latin
fr						Latin
zh						Han
ru						Cyrillic
hi						Devan.
ar						Arabic

Table 2: **Meaning-matched pairs** per language and split, share passing the entailment filter (QC%), and agreement (κ) between the filter and a native-speaker audit of 100 pairs.

Prompt sets. Table 3 lists the sources and sizes of the three prompt sets: English requests without a language cue (NO-CUE), language-neutral inputs (NEUTRAL) and requests with an explicit language intent (INTENT). NO-CUE prompts are shared across target languages, so the same English request is evaluated under each language's edit. Splits are disjoint at the level of the source item.

Cue-strength prompts. For test D6 (Appendix F) each of 100 NO-CUE items is rewritten at six cue levels, giving 600 cue conditions per model–language pair, with the content held fixed: L0 no cue; L1 a topic tied to a region where the language is spoken; L2 one target-language word in the prompt; L3 an English instruction to answer in the target language; L4 the same instruction written in the target language; L5 the whole prompt in the target language.

Romanized variants. For the script control (Appendix H.1) we use romanized Hindi from the Dakshina dataset (Roark et al., 2020) and romanize Russian with ISO 9 (International Organization for Standardization, 1995).

Condition	Definition	Source and license	$\mathcal{D}_{\text{disc}}$	$\mathcal{D}_{\text{dev}}$	$\mathcal{D}_{\text{eval}}$	Example
NO-CUE	English instruction with no mention of any language	**[source; license]**				**[example prompt]**
NEUTRAL	Arithmetic, code, lists of names, symbols	**[source; license]**	–			**[example prompt]**
INTENT	Asks for English, or written natively in another suite language	**[source; license]**	–			**[example prompt]**

Table 3: **Prompt sets.** Counts are prompts per target language. NO-CUE covers seven task types in equal proportion: question answering, explanation, summarization, writing, advice, coding and reasoning.

B Models, Dictionaries and Dictionary Quality

Exactness of the null edit. For every model we ran 1,000 prompts with the hook installed at $\gamma = 0$ and without it. The maximum absolute logit difference is 0.000000 and the greedy outputs are identical for 100% of prompts, for all four models.

Dictionary quality on instruction-tuned activations. Table 5 reports, for the layers inside each model's window, the fraction of variance unexplained (FVU), the mean number of active features per token (L0), the share of features never active on $\mathcal{D}_{\text{dev}}$ prompts, and the increase in next-token cross-entropy when the reconstruction replaces the activation (ΔCE). FVU stays between 0.27 and 0.39 for every model, below the 0.5 level at which we would expect the dictionaries to be unusable. Gemma-2-2B-it has the weakest dictionaries (FVU 0.391, 29.4% dead features, ΔCE 0.18 nats), consistent with its weaker steering results. Because the edit is added to the residual stream and never replaces it, reconstruction error does not enter the forward pass.

Dictionaries trained on instruction-tuned models. Gemma Scope also releases dictionaries trained on Gemma-2-9B-it for a few layers (Lieberum et al., 2024). At such a layer inside the Gemma-2-9B-it window (layer **[layer]**), we build a single-layer Neural FOXP2 edit twice, once with each dictionary and with the same Stage I–III rules and γ tuning, and evaluate both on the NO-CUE evaluation prompts (Table 6). Both rows come from the single-layer run, not from the full multi-layer edit. The instruction-trained dictionary changes DEFAULT by -0.013 (95% CI $[-0.031, +0.006]$), the full-response rate by -0.8 points ($[-3.1, +1.4]$) and utility by $+0.06$ points ($[-0.08, +0.19]$); all three intervals include zero.

Model	Checkpoint (revision)	SAE release (revision)	SAE trained on	Layers	Width m	Hook
Llama-3.1-8B-Instruct	[hash]	Llama Scope [rev.]	base			residual
Gemma-2-9B-it	[hash]	Gemma Scope [rev.]	base†			residual
Gemma-2-2B-it	[hash]	Gemma Scope [rev.]	base			residual
Qwen3-8B	[hash]	Qwen-Scope [rev.]	base			residual

Table 4: **Models and SAE dictionaries.** All three suites were trained on base-model activations (He et al., 2024; Lieberum et al., 2024; Deng et al., 2026); †Gemma Scope additionally releases dictionaries trained on Gemma-2-9B-it for a few layers. Revisions are pinned; feature indices are only meaningful for the listed revision.

Model	FVU↓	L0	Dead %↓	ΔCE↓
Llama-3.1-8B-Instruct	0.312	68.4	21.3	0.12
Gemma-2-9B-it	0.274	74.8	17.6	0.09
Gemma-2-2B-it	0.391	61.7	29.4	0.18
Qwen3-8B	0.337	82.1	23.7	0.14

Table 5: **Dictionary quality** on instruction-tuned activations, averaged over window layers. ΔCE: increase in next-token cross-entropy (nats) when the SAE reconstruction replaces the activation. Per-layer values are in the released results file.

Dictionary	DEFAULT↑	Full resp. %↑	ΔS
Base-trained	[?]	[?]	[?]
Instruction-trained	[?]	[?]	[?]
Difference	−0.013	−0.8	+0.06

Table 6: **Base- vs instruction-trained dictionaries** for a single-layer edit in Gemma-2-9B-it at layer [layer], macro over six languages. Both rows must come from the single-layer run and are being recomputed; the differences are paired estimates from that run. Difference: instruction-trained minus base-trained; paired bootstrap 95% CIs $[-0.031, +0.006]$, $[-3.1, +1.4]$ and $[-0.08, +0.19]$.

C Method Details

Effective rank and eigengap. With singular values $\sigma_1 \geq \sigma_2 \geq \dots$ of $\Delta Z^{(\ell)}$ and $p_i = \sigma_i / \sum_k \sigma_k$, the effective rank is $r_{\text{eff}} = \exp(-\sum_i p_i \log p_i)$ and the eigengap index is $i^\star = \arg\max_{i \leq i_{\max}} \sigma_i/(\sigma_{i+1} + \epsilon)$. We set $r_\ell = \min(\lceil r_{\text{eff}} \rceil, i^\star)$.

Subspace stability. For B bootstrap resamples of the discovery pairs we recompute the rank-r_ℓ subspace $\mathcal{S}^{(\ell,b)}$ and measure its agreement with the full-sample subspace through principal angles $\angle_i$:

$$\text{Stab}^{(\ell)} = 1 - \frac{1}{B}\sum_{b=1}^{B}\frac{1}{r_\ell}\sum_{i=1}^{r_\ell}\sin\angle_i\big(\mathcal{S}^{(\ell,b)}, \mathcal{S}^{(\ell)}\big).$$

Layers with $\text{Stab}^{(\ell)} < \tau_{\text{stab}}$ are not eligible for the window.

Window objective. Among contiguous ranges W of eligible layers with width in $[w_{\min}, w_{\max}]$, we select

$$\mathcal{W} = \arg\max_{W} \text{Gain}(W) - \alpha_{\text{leak}}\,\text{Leak}(W) - \alpha_{\text{kl}}\,\text{KL}(W) - \alpha_{\text{util}}\,|\Delta S(W)|,$$

subject to the guardrails $\text{Leak} \leq \epsilon_{\text{leak}}$, $\text{KL} \leq \epsilon_{\text{KL}}$ and $|\Delta S| \leq \epsilon_{\text{util}}$, all measured on $\mathcal{D}_{\text{dev}}$.

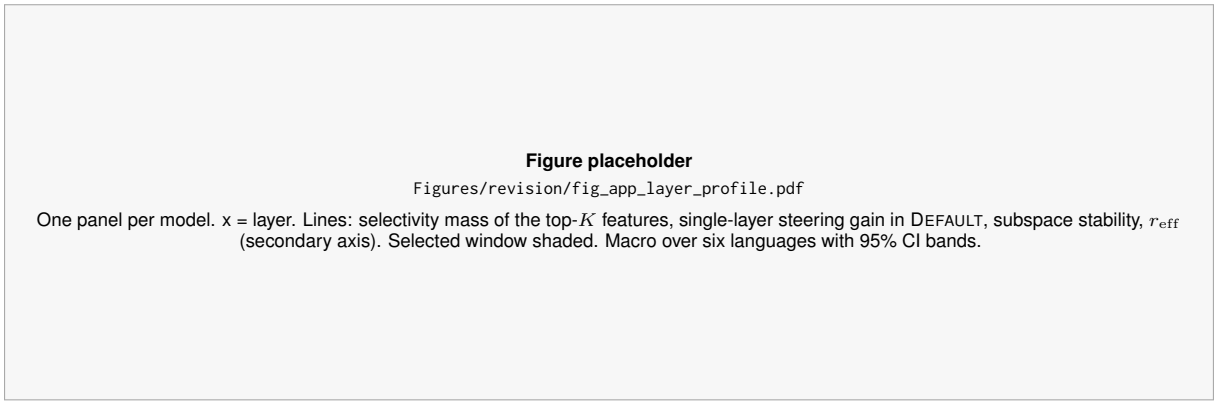


Figure 5: **Layer profiles.** Language selectivity, single-layer steering effect, subspace stability and effective rank by layer; the selected window is shaded.

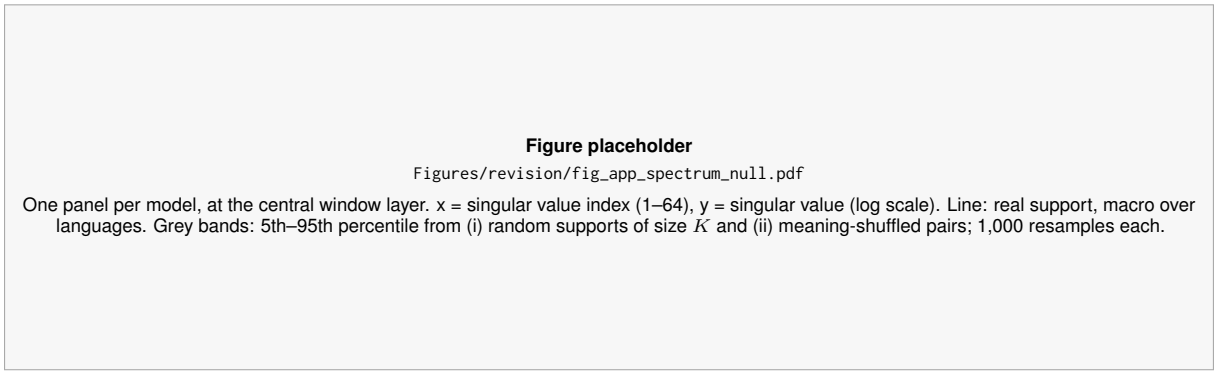


Figure 6: **Low-rank structure against null models.** Singular value spectra of the language-shift matrix compared with random feature supports and meaning-shuffled pairs.

Model	Effective rank			Var. (%)
	Real	Null A	Null B	
Llama-3.1-8B	8.7	12.9	18.6	78.4
Gemma-2-9B	9.4	13.7	19.8	75.9
Gemma-2-2B	7.2	11.8	17.1	81.6
Qwen3-8B	8.1	12.5	18.2	79.3
Macro	8.4	12.8	18.4	78.8
Pairs, median	8.3	12.7	18.5	79.0
Pairs, IQR	7.1–9.5	11.6–13.8	17.0–20.1	

Table 7: **Low-rank structure against null models** at the central window layer. Null A: random feature supports of size K; Null B: meaning-shuffled pairs. Var. top-r: share of variance explained by the top r singular directions of the real matrix (67.2% and 55.4% for Nulls A and B across pairs). Model rows: macro over languages at the central window layer, nulls as median over resamples. Pairs: per-pair median over the layers of the selected window, summarized over the 24 model–language pairs (a different population from the layer-level values of Table 9); for 20 of 24 pairs the real effective rank is below the 5th percentile of Null B.

Parameter	Value	Role
K	64	features kept per layer (from 16, 32, 64, 128)
$\alpha_{1,2,3}$		lift probe magnitudes
r_ℓ rule	$\min(\lceil r_{\text{eff}} \rceil, i^\star)$	subspace rank
B, τ_{stab}	200, 0.70	stability bootstrap, threshold
$w_{\min}$, $w_{\max}$	15%, 35%	window width bounds (share of model depth)
β/λ	1.0	suppression relative to promotion
γ	per pair	scale, from 0.25, 0.5, 0.75, 1, 1.5, 2, 3; median 0.75 (Table 9)
T	$\{1, 2, 3\}$	metric horizon (D2)
ϵ_{leak}, ϵ_{KL}, ϵ_{util}		guardrails
Seeds		splits and sampling
Frozen on	**[date]**	before any $\mathcal{D}_{\text{eval}}$ run

Table 8: **Operating point.** All values fixed on $\mathcal{D}_{\text{dev}}$ and shared by all models and languages unless a sweep varies them; γ is selected per pair. With $\gamma = 0$ the maximum logit difference from the unedited model is 0.0000.

Model	Lang.	Window	r	r_{eff}	γ	Constr.
Llama-3.1-8B	es	10–18	6	8.9	0.75	
	fr	9–17	5	8.1	0.75	
	zh	11–20	7	10.3	1.5	
	ru	9–18	6	9.2	1	
	hi	12–20	5	7.8	0.5	
	ar	8–16	5	8.4	0.5	
Gemma-2-9B	es	12–24	7	10.2	1	
	fr	13–25	6	9.5	0.75	
	zh	14–27	8	11.4	1.5	
	ru	12–24	7	10.6	1	
	hi	15–28	6	8.8	0.25	yes
	ar	11–23	6	9.0	0.75	
Gemma-2-2B	es	7–14	5	7.1	0.75	
	fr	6–13	4	6.8	0.5	
	zh	8–15	6	8.4	1.5	
	ru	7–14	5	7.4	1	
	hi	8–16	4	6.2	0.25	yes
	ar	6–13	4	6.5	0.5	
Qwen3-8B	es	10–20	6	8.7	1	
	fr	11–21	6	8.2	0.75	
	zh	12–22	7	10.1	1.5	
	ru	10–20	6	9.0	1	
	hi	13–23	5	7.6	0.25	yes
	ar	9–19	5	7.8	0.75	

Table 9: **Artifacts per model and language**: selected window $\mathcal{W}$ (layers), subspace rank $r = \min(\lceil r_{\text{eff}} \rceil, i^\star)$, raw effective rank r_{eff}, and scale γ at the operating point. r_{eff} here summarizes the layer-level distribution across the layers of the selected window (median 8.55, IQR 7.75–9.27 over pairs), a different summary from the per-pair medians of Table 7 (8.3, IQR 7.1–9.5). Windows span 28–35% of model depth. Constr.: pair flagged as constrained, i.e. not meeting every guardrail on $\mathcal{D}_{\text{dev}}$ (3 of 24, all Hindi).

D Baseline Implementations

All baselines use the same prompts, decoding settings, metrics and guardrails as Neural FOXP2. Steering baselines are applied in the same layer window as Neural FOXP2 unless stated otherwise, and their strength is tuned on $\mathcal{D}_{\text{dev}}$ under the guardrails of Table 8; Figure 3 additionally reports their full strength sweeps.

Prompting. The *short instruction* appends "Answer in *X*." to the user turn. The *system prompt* adds the fixed system message "Always respond in *X*."; it is the same for every model and language and is not tuned. The *best prompt* is selected separately for every model–language pair: each of the ten strategies in Table 10 is evaluated on $\mathcal{D}_{\text{dev}}$ and the one with the highest DEFAULT is kept. The system message is strategy P3, so the best prompt can never be worse than the system prompt on $\mathcal{D}_{\text{dev}}$; on $\mathcal{D}_{\text{eval}}$ it is 1.8 points better (Table 1). Table 11 lists the tuning choices and development budget of every method.

Difference-in-means. At each layer in $\mathcal{W}$ we add $\gamma_4\, v^{(\ell)}$ with $v^{(\ell)} = \mathbb{E}_k[h^{(\ell)}(x^{(k)}_{\ell_t}) - h^{(\ell)}(x^{(k)}_{\text{en}})]$, in the style of contrastive activation addition (Rimsky et al., 2024). The scale γ_4 is tuned per model–language pair on $\mathcal{D}_{\text{dev}}$ under the same guardrails as Neural FOXP2; the median selected γ is 0.75 (Table 11).

Difference-in-means with projection. Difference-in-means steering plus removal of the component of $h^{(\ell)}$ along the mean English-prompt direction, analogous to directional ablation (Arditi et al., 2024). It is the dense counterpart of Eq. 6 without SAE support or subspace.

LAPE neurons. Language activation probability entropy (Tang et al., 2024) selects FFN neurons whose activation probability concentrates on the target language; we add a mean target-minus-English shift on those neurons, with the number of edited neurons per layer matched to K.

SAE features. Following Chou et al. (2025), we add the decoder directions of the top-n features ranked by selectivity alone, without causal lift, subspace or window search; $n \in \{1, 4, 16\}$ and the strength are tuned per pair on $\mathcal{D}_{\text{dev}}$ (n=1 for 4 pairs, n=4 for 8, n=16 for 12).

LoRA. For each model and language we finetune LoRA adapters (Hu et al., 2022) (rank **[16]**, α = **[32]**, learning rate **[2e-4]**; 1–3 epochs, selected per pair by $\mathcal{D}_{\text{dev}}$ full-response rate: 1 epoch for 8 pairs, 2 for 11, 3 for 5) on the $\mathcal{D}_{\text{disc}}$ NO-CUE prompts paired with the model's own answers generated under an explicit instruction to use the target language and filtered by language identification. No evaluation prompt is used in training.

Strategy	Template	$\Delta M\uparrow$	DEFAULT↑	Full resp. %↑	ΔS	Sel.
P1 Instruction before request	"Answer in *X*." + request					5
P2 Instruction after request	request + "Answer in *X*."					4
P3 System message	system: "Always respond in *X*."					3
P4 Instruction in target language	instruction written in *X*					3
P5 Paraphrase average	three paraphrases of P2					2
P6 One-shot exemplar	one request–answer pair in *X*					2
P7 Three-shot exemplars	three request–answer pairs in *X*					2
P8 Script cue	P2 + a greeting in *X*					1
P9 Persona	system: "You are an assistant for *X* speakers."					1
P10 Combined	P3 + P4 + P8					1

Table 10: **Prompt strategies**, macro over 24 model–language pairs on $\mathcal{D}_{\mathrm{dev}}$. The best-prompt baseline uses the best strategy per pair; Sel. counts the pairs (of 24) for which each strategy was selected. P3 is the system-prompt baseline.

				$\mathcal{D}_{\mathrm{dev}}$ evaluations	
Method	**Tuned**	**Selection**	**Guardrails met**	**per pair**	**total**
Short instruction	–	fixed instruction	–	200	4,800
Best prompt	strategy	10 strategies per pair	–	2,000	48,000
System prompt	–	fixed system message	–	200	4,800
Difference-in-means	γ	median $\gamma = 0.75$	19/24	1,400	33,600
DiM + projection	γ	median $\gamma = 1.0$	21/24	1,400	33,600
LAPE neurons	γ	median $\gamma = 0.75$	22/24	1,400	33,600
SAE features	n, γ	n=1/4/16: 4/8/12 pairs; median $\gamma = 0.75$	17/24	4,200	100,800
LoRA	epoch	1/2/3 epochs: 8/11/5 pairs	–	$\leq$600	$\leq$14,400
Neural FOXP2	γ	median $\gamma = 0.75$	21/24	1,400	33,600
Neural FOXP2 + system prompt	–	reuses the Neural FOXP2 artifact and γ	–	200	4,800

Table 11: **Tuning and development budget.** Tuned parameter, selected settings over the 24 pairs, number of pairs meeting every guardrail at the selected setting, and number of candidate evaluations on $\mathcal{D}_{\mathrm{dev}}$. LoRA selects its epoch by $\mathcal{D}_{\mathrm{dev}}$ full-response rate and is not subject to the guardrails.

E Metric Definitions

Token sets. For each tokenizer, V_{tgt} contains tokens that occur in target-language FLORES text at least [ρ] times more often than in English text after normalization, and V_{en} is built symmetrically. A shared pool V_{shared} holds digits, punctuation, whitespace, URL fragments and tokens frequent in both languages; these receive $w(u) = 0$, all others $w(u) = 1$. Token sets are pinned to the tokenizer hash. Alternative constructions are compared in test D10.

DEFAULT. A prompt counts as a success when $\Delta M > 0$ and GlotLID (Kargaran et al., 2023) assigns the target language to the first [20] generated tokens; fastText `lid.176` (Joulin et al., 2017) is used as a second identifier, and agreement between the two is reported in Table 18.

Full-response rate and switch-away rate. The response is split into sentences; code blocks are removed. A response is *in the target language* when every sentence of at least 20 characters is identified as the target language. The switch-away rate is the share of responses whose first sentence is in the target language and some later sentence is not.

LCB pass rate. Line-level pass rate as defined by Marchisio et al. (2024), computed with the benchmark's code on our outputs.

Leakage. For an edit toward ℓ_t, leakage is the mean absolute increase in off-target DEFAULT, with DEFAULT expressed as a fraction: $\frac{1}{5}\sum_{\ell' \neq \ell_t} \max\big(0, \text{DEFAULT}_{\ell'}(\text{edit}) - \text{DEFAULT}_{\ell'}(\text{none})\big)$ over the other five suite languages. Every table uses this definition and unit; Figure 4 shows the same changes in percentage points, and its off-diagonal mean, 0.65 points, is the Neural FOXP2 leakage of 0.0065 in Table 1. Leakage values for the other methods, ablations and sweeps are being recomputed from the raw matrices under this definition.

KL drift. $\frac{1}{|T|}\sum_{t\in T} \mathrm{KL}\big(p_{\text{edit}}(\cdot \mid x, y_{<t}) \,\|\, p_\theta(\cdot \mid x, y_{<t})\big)$, averaged over NO-CUE prompts.

Utility. Regime A: XNLI (Conneau et al., 2018) and Belebele (Bandarkar et al., 2024) accuracy by log-likelihood over label options; XL-Sum (Hasan et al., 2021) ROUGE-L with a multilingual tokenizer; FLORES en→X, scored with chrF++ (Popović, 2017) and COMET (Rei et al., 2020) and reported as a composite of the two ([**how the two are combined**]). Regime B: accuracy on an MMLU subset (Hendrycks et al., 2021) (option letter by log-likelihood) and GSM8K (Cobbe et al., 2021) (final number extracted independently of the output language). Regime C: an LLM judge rates fluency and adequacy of NO-CUE answers on a 1–5 scale with the rubric in Appendix K. ΔS for a method is the predeclared mean of its changes against the unedited model on the six benchmarks of regimes A and B (XNLI, Belebele, XL-Sum, FLORES, MMLU and GSM8K). Results are reported per benchmark, with the six-benchmark aggregate in the main comparison; FLORES chrF++ and COMET are reported individually in Table 41, and the regime C ratings are reported separately.

F Validation of the Defaultness Score

Each test below is run on all 24 model–language pairs; aggregates are macro averages with bootstrap intervals.

F.1 Forms and horizon (D1, D2)

Besides ΔM (Eq. 2) we consider the log-odds form and the target-share form,

$$\Delta L(x) = \frac{1}{|T|}\sum_{t\in T} \log \frac{M_{\text{tgt}}(x,t)+\epsilon}{M_{\text{en}}(x,t)+\epsilon},$$

$$S(x) = \frac{1}{|T|}\sum_{t\in T} \frac{M_{\text{tgt}}(x,t)}{M_{\text{tgt}}(x,t)+M_{\text{en}}(x,t)+\epsilon},$$

and a decayed weighting $1/t$ in place of the uniform average. **D1** selects the form with the highest AUC for full-response language on $\mathcal{D}_{\text{dev}}$. **D2** computes the same AUC for $T = \{1,\dots,k\}$, $k = 1,\dots,10$, and selects the smallest k within 0.01 of the maximum.

Table 12 gives the $\mathcal{D}_{\text{dev}}$ results. The rule selects k=3 for ΔM and S, k=5 for the $1/t$-weighted ΔM and for ΔL, and k=10 for prefix language identification. S is at or above ΔM at every horizon on $\mathcal{D}_{\text{dev}}$ (by at most .013), so D1 selects S, and S is the primary predictive metric of the paper; ΔM is kept as the mechanistically interpretable metric used by Stage I and DEFAULT (Section 3.2). The two remain within each other's 95% intervals at every k on $\mathcal{D}_{\text{eval}}$ (pooled prompt-level AUC; Figure 7a). The decayed weighting never improves on the uniform average.

To write once results are in: Panel (b) still needs the reliability-bin data (10 bins of ΔM vs observed full-response target rate, per model).

Form	k=1	k=2	k=3	k=5	k=10	Sel. k
ΔM (uniform)	.778	.817	.843	.843	**.851**	3
ΔM $(1/t)$	.774	.808	.835	.841	**.846**	5
ΔL (uniform)	.767	.803	.829	.839	**.841**	5
S (uniform)	.785	.830	.848	.850	**.851**	3
Prefix LID (ref.)	.746	.784	.816	.844	**.860**	10

Table 12: **D1–D2: form and horizon selection.** AUC for full-response language on $\mathcal{D}_{\text{dev}}$, macro over 24 pairs; bold marks each row's maximum. Sel. k: smallest horizon within 0.01 of the row maximum.

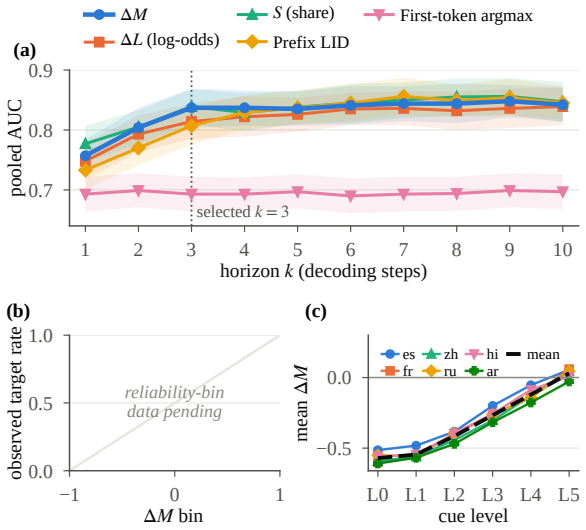


Figure 7: **Validity of the defaultness score.** (a) Pooled prompt-level AUC for the language of the finished response on $\mathcal{D}_{\text{eval}}$ against the horizon k (not the macro AUC over pairs of Table 13: .837 pooled vs .819 macro for ΔM at k=3), with 95% intervals (shaded); the dotted line marks the horizon selected on $\mathcal{D}_{\text{dev}}$. (b) Calibration against the finished response. (c) Mean ΔM at six cue levels, from no cue (L0) to a prompt written entirely in the target language (L5). Macro over four models.

F.2 Choice of the horizon

The horizon T decides what the defaultness score is allowed to see, and it was fixed to $T = \{1, 2, 3\}$ on $\mathcal{D}_{\text{dev}}$ by the rule D2 before any evaluation run. This subsection collects the evidence behind that choice and explains why neither a shorter nor a longer horizon would serve the score better. It is organized around four questions: whether more steps make the score more accurate, whether the model has committed to a language by the third step, what happens to steered responses after the third step, and what longer horizons cost. Note that the choice of T affects only ΔM and the quantities derived from it (DEFAULT, KL drift and the Stage I ranking); the full-response rate, our primary outcome, is computed from the finished text and does not depend on T at all.

Does accuracy keep improving with more steps? It does not. Figure 8a plots, for every form of the score, the gap between the AUC at horizon k and the best AUC of that form on $\mathcal{D}_{\text{dev}}$. The uniform ΔM is within the tolerance of 0.01 at k=3 (.843 against a maximum of .851 at k=10), so the rule selects three steps. The target-share form S is also within tolerance at k=3 (.848 against .851). The $1/t$-weighted ΔM and the log-odds form ΔL converge more slowly and are selected at k=5, but neither exceeds the uniform ΔM at its own selected horizon (.841 and .839, against .843 at k=3), so a longer horizon would buy a more complicated score without a more accurate one. Prefix language identification behaves differently: on the same three tokens it reaches only .816 and needs ten tokens to reach its maximum (.860). Because ΔM reads the whole next-token distribution rather than the single token that is emitted, it extracts more information from each step, which is why it saturates earlier.

Figure 8b shows the same pattern on $\mathcal{D}_{\text{eval}}$, in pooled prompt-level AUC, as the change in AUC contributed by each additional step. For ΔM, the second and third steps add .047 and .033; every later step changes the AUC by between $-.006$ and $+.006$, and the seven steps from k=3 to k=10 add .005 in total, well inside the 95% intervals of Figure 7a. The first step alone is not enough either: ΔM at k=1 trails k=3 by .065 on $\mathcal{D}_{\text{dev}}$ and .080 on $\mathcal{D}_{\text{eval}}$, and the first-token argmax, which also uses a single step, stays near .69–.70 on $\mathcal{D}_{\text{eval}}$ whatever the horizon. The first token can be a formatting mark, a digit or the beginning of a named entity,

which carry little information about the language that follows (Appendix F.10).

Has the model committed by the third step? In all four models it has, for the most part. Figure 9 decomposes the commitment curves of Figure 16. With no forced tokens, the share of responses that stay in the target language is the unedited full-response rate, 1.9% (Table 1). The first forced target-language token raises it by .490 on average, the second by .168 and the third by .097; from three to five tokens the gain falls to .027 per token, and from five to ten tokens to .008 per token (panel a). Measured against the effect of forcing ten tokens, three tokens already deliver 89% of it on the mean curve (panel b). Per-model values at $t{=}0$ must equal each model's Experiment 1 rate and are being regenerated, so per-model curves are shown from $t{=}1$. The same is true in the opposite direction: with three English tokens forced, 97.3% of responses stay in English to the end (Appendix G.1). By the third step, then, the next-token distributions largely reflect a language that has already been chosen, and extending the horizon would add steps that describe the continuation of that choice rather than the choice itself.

What happens after the third step under steering? Under Neural FOXP2, the share of responses whose first k tokens are in the target language is 49.8% at $k{=}1$, 64.1% at $k{=}2$ and peaks at 74.6% at $k{=}3$; it then declines to 72.8% at five tokens, 69.4% at ten and 58.7% over the full response (Figure 20). The edit is thus strongest on the opening of the response, which is exactly the part of decoding the score reads, and the later decline comes from responses that start in the target language and then switch away. Switching away is a distinct failure with its own measures, the full-response rate and the switch-away rate (Appendix E). A longer horizon would mix the two phenomena in one number and make ΔM a blunter objective for the Stage I search and the window selection, which need to reward edits that change the decision.

What does a longer horizon cost? Computing ΔM at horizon k requires k decoding steps per prompt, and ΔM is evaluated many times while an artifact is built: for every candidate feature at three probe strengths in the Stage I lift, and for every candidate window and scale on $\mathcal{D}_{\text{dev}}$; KL drift is measured over the same steps. A ten-step horizon would multiply the scoring cost of each of these computations by $10/3 \approx 3.3$ with, as shown above, no measurable gain in validity. Figure 10 puts the four considerations side by side. Prediction and commitment both reach 88–89% of their maximum at three steps, the steered opening reaches its peak there, and the cost of scoring keeps growing linearly beyond it. Three steps is the shortest horizon at which all three signals are close to their maximum.

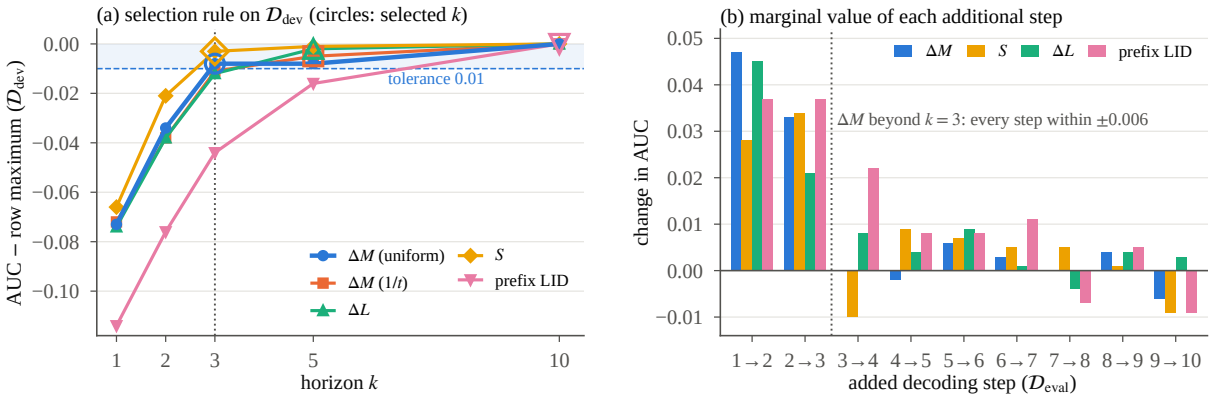


Figure 8: **Selecting the horizon.** (a) Development-set AUC of each form of the score minus that form's maximum over $k \in \{1, 2, 3, 5, 10\}$ (Table 12); the shaded band is the 0.01 tolerance of rule D2, and open circles mark the horizon each form selects. ΔM and S select k=3, the $1/t$-weighted ΔM and ΔL select k=5, and prefix language identification needs k=10. (b) Change in AUC on $\mathcal{D}_{\text{eval}}$ contributed by each additional decoding step (Figure 7a). After the third step the contributions to ΔM are indistinguishable from zero.

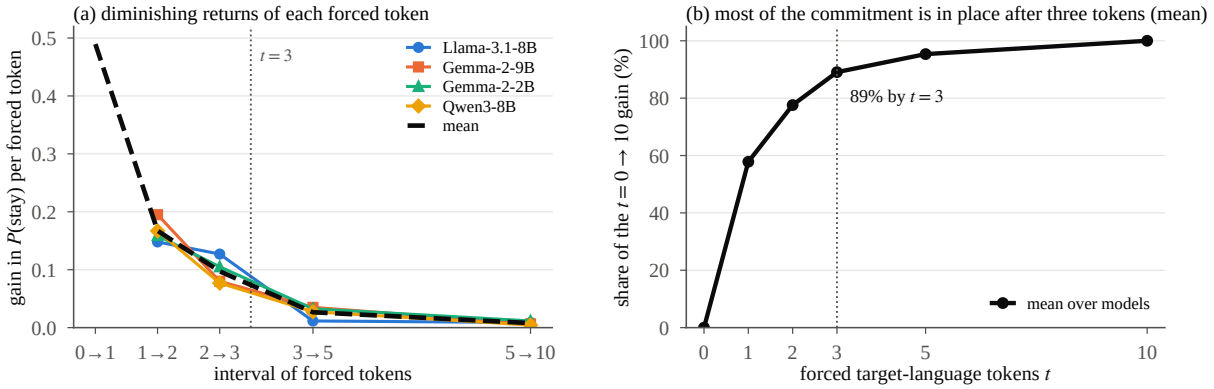


Figure 9: **Commitment saturates after about three tokens.** (a) Gain in the probability that an unedited response stays in the target language to the end, per additional forced target-language token, for each model (from t=1) and for the mean over models (from t=0, where it equals the unedited full-response rate of 1.9%); gains for the intervals 3–5 and 5–10 are divided by the number of tokens added. (b) Share of the increase of the mean curve from t=0 to t=10 that is reached after t forced tokens. Data as in Figure 16.

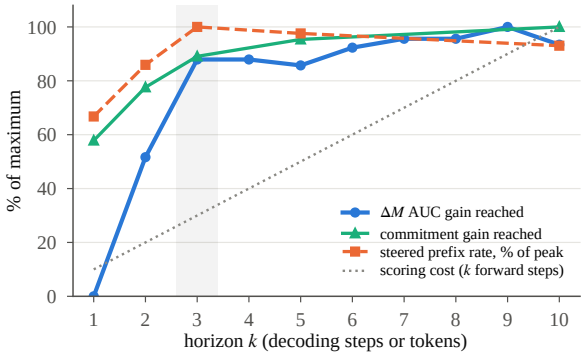


Figure 10: **Evidence and cost against the horizon.** Blue: share of the $\mathcal{D}_{\text{eval}}$ AUC gain of ΔM (from k=1 to its maximum) reached at k. Green: share of the commitment gain (from t=0 to t=10, mean over models) reached at t=k. Orange: share of Neural FOXP2 responses whose first k tokens are in the target language, relative to its peak. Dotted: relative cost of computing ΔM, proportional to k. The shaded column marks the selected horizon k=3.

F.3 Criterion and comparative validity (D3, D4)

Predictor	$\rho\uparrow$	AUC$\uparrow$	AUC$_{\text{h}}\uparrow$	Cost
ΔM (ours)	.796	.819	.814	3 steps
Log-odds form			.802	3 steps
Target-share form S	[?]	[?]	.827	3 steps
First-token argmax		.733	.706	1 step
Prefix LID, 3 tokens		.787	.779	3 steps
Prefix LID, 20 tokens		.823	.842	20 steps
Perplexity ratio		.724	.731	2 passes
LLM judge		.834	.851	full output

Table 13: **Predicting the language of the finished response**, macro AUC over 24 model–language pairs ($\mathcal{D}_{\text{eval}}$); the pooled prompt-level AUC of Figure 7a is a different aggregation. S is the primary predictive metric, ΔM the mechanistically interpretable one; red cells are pending recomputation under the macro aggregation. ρ: Spearman correlation with the full-response label (mean over models); AUC: against the full-response label; AUC$_{\text{h}}$: against human language judgments (Section 6.6). Per-pair spread and per-model correlations are in Table 14.

D3 correlates per-prompt ΔM with the full-response language label, with human language labels (Appendix K) and with LLM-judge labels. **D4** compares ΔM by AUC with first-token argmax language, prefix language identification at 3 and 20 tokens, the perplexity ratio between target and English continuations, and an LLM judge (Table 13, Figure 11).

Table 14 gives the per-pair spread and the per-model correlations. ΔM correlates strongly with the full-response language label (Spearman .772–.855) and with LLM-judge appropriateness (.601–.788), and only weakly with the downstream task score (.152–.271), so it tracks the language of the response rather than its quality. AUCs are computed on the cue-strength prompts, where both outcome classes occur (Table 15); the target-language share there ranges from 9.4% to 82.0% across pairs.

(a) AUC over 24 model–language pairs

Predictor	Mean	SD	Mean 95% CI
ΔM (k=3)	.819	.049	[.788, .850]
Prefix LID, 3 tok.	.787	.042	[.755, .818]
Prefix LID, 20 tok.	.823	.042	[.792, .854]
First-token argmax	.733	.039	[.701, .764]
Perplexity ratio	.724	.036	[.693, .755]
LLM judge	.834	.043	[.803, .865]

(b) Correlation of ΔM with external criteria

	Full-resp. LID			LLM judge			Task score		
Model	r	ρ	n	r	ρ	n	r	ρ	n
L-8B	.771	.855	591	.708	.788	598	.311	.202	600
G-9B	.828	.772	578	.680	.601	550	.231	.152	569
G-2B	.822	.786	556	.687	.714	566	.368	.175	582
Q-8B	.906	.772	569	.635	.686	596	.281	.271	581

Table 14: **D3–D4: criterion and comparative validity** on $\mathcal{D}_{\text{eval}}$. (a) Mean, standard deviation and mean 95% interval bounds of the per-pair AUC for predicting full-response language. (b) Pearson r and Spearman ρ between per-prompt ΔM and each criterion; the LLM judge rates whether the response language is appropriate, and the task score is the downstream quality score of the response.

Model	es	fr	zh	ru	hi	ar
Llama-3.1-8B-Instruct	51.4 (542)	28.2 (592)	56.1 (558)	35.4 (556)	39.5 (591)	15.7 (557)
Gemma-2-9B-it	13.7 (592)	73.6 (562)	78.9 (560)	25.5 (577)	19.1 (584)	22.9 (557)
Gemma-2-2B-it	69.4 (587)	11.9 (555)	22.5 (575)	45.5 (563)	25.8 (592)	9.4 (583)
Qwen3-8B	82.0 (580)	63.9 (574)	56.2 (571)	67.8 (565)	34.0 (556)	52.6 (584)

Table 15: **Evaluation pool for D1–D5.** Share (%) of cue-pool responses whose full response is in the target language, with the number of responses whose language could be determined in parentheses.

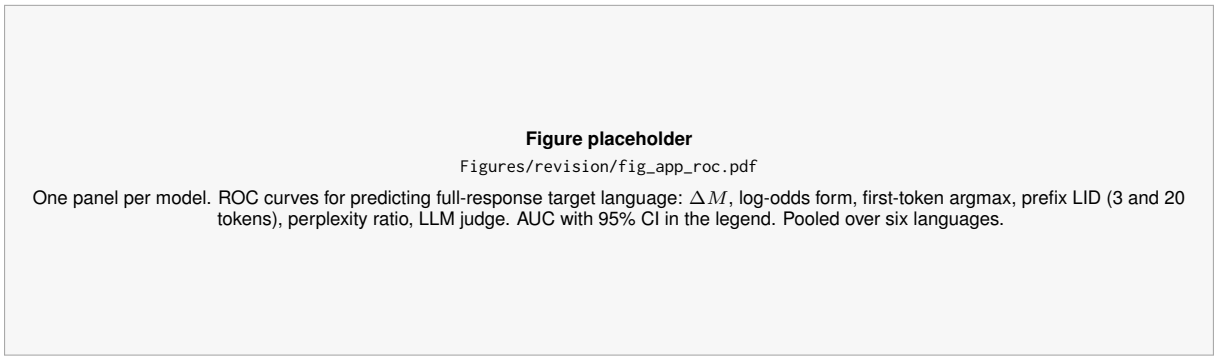


Figure 11: **D4: predicting the finished response.** ROC curves of the defaultness score and alternative predictors.

F.4 Calibration (D5)

We fit an isotonic map from ΔM to the probability that the full response is in the target language on $\mathcal{D}_{\text{dev}}$ for each model and report expected calibration error (Guo et al., 2017) on $\mathcal{D}_{\text{eval}}$, both for the model's own map and for maps transferred from each other model. Own maps reach .048–.071 (mean .060); transferred maps reach .076–.110 (mean .091). Calibration is therefore model-specific, and any probability reading of ΔM uses the model's own map.

	Evaluated on			
Map fitted on	**L-8B**	**G-9B**	**G-2B**	**Q-8B**
Llama-3.1-8B-Instruct	**.048**	.081	.093	.098
Gemma-2-9B-it	.076	**.071**	.088	.095
Gemma-2-2B-it	.096	.083	**.061**	.110
Qwen3-8B	.083	.087	.100	**.059**

Table 16: **D5: calibration.** Expected calibration error (lower is better) of the map from ΔM to full-response probability. Rows: model whose $\mathcal{D}_{\text{dev}}$ data the map was fitted on; columns: model it is evaluated on. The diagonal (bold) uses the model's own map and is the lowest entry in every row and column.

F.5 Known-groups validity (D6)

ΔM should increase with the strength of the language cue (levels L0–L5, Appendix A). We test the ordered alternative with the Jonckheere–Terpstra test (Jonckheere, 1954) per model and language and report the macro result in Figure 7c. Averaged over models and languages, ΔM rises from -0.571 (L0) through -0.545, -0.415, -0.267 and -0.121 to 0.030 (L5). It increases at every level for five languages; for Russian, the regional-topic cue (L1) leaves it unchanged (-0.550 to -0.557). Only L5 moves the score above zero on average, and for Arabic it stays just below zero even there (-0.028). **[Jonckheere–Terpstra statistic and p, macro and per model.]**

F.6 Reliability and discriminant validity (D7, D8)

D7 computes the intraclass correlation ICC(2,1) (Shrout and Fleiss, 1979) of ΔM across five paraphrases of each prompt. Unedited NO-CUE responses are almost all English, so their ΔM varies little and an ICC on them would be low only through restricted range; we therefore use a separate subset of 50 items drawn from the cue-strength pool, each paraphrased five times and rendered at all six cue levels, with item-at-level as the subject. Split-half reliability splits V_{tgt} and V_{en} at random into halves 100 times and reports the Spearman–Brown corrected correlation between the two halves' ΔM. **D8** computes partial Spearman correlations of ΔM with properties it should not track (prompt length, response length and answer adequacy judged as in regime C), controlling for the share of target-language sentences, and the correlation ratio η of ΔM over the seven task types.

ΔM is reliable: ICC is 0.71–0.82 across models (mean 0.77) and split-half reliability 0.85–0.92 (mean 0.89), with every lower 95% bound at or above 0.63 and 0.80 respectively (Table 17). It is nearly unrelated to prompt length ($\rho \leq 0.06$) and task type ($\eta \leq 0.08$), and only weakly related to answer adequacy ($\rho \leq 0.12$). The largest association is with response length ($\rho = 0.09$–0.19, mean 0.14), possibly because longer responses give more room to switch away from the opening language.

Model	ICC	Split	ρ_{prompt}	η_{task}	ρ_{resp}	ρ_{adeq}
L-8B	0.79	0.91	+0.06	0.04	+0.12	+0.08
G-9B	0.75	0.88	+0.03	0.06	+0.17	+0.10
G-2B	0.71	0.85	−0.02	0.08	+0.19	+0.12
Q-8B	0.82	0.92	+0.05	0.03	+0.09	+0.07
Mean	0.77	0.89	+0.03	0.05	+0.14	+0.09

Table 17: **D7–D8: reliability and discriminant validity**, macro over languages. ICC(2,1) across paraphrases on the cue pool; Split: Spearman–Brown corrected split-half reliability over token-set halves; partial Spearman ρ of ΔM with prompt length, response length and answer adequacy, controlling for the target-sentence share; η: correlation ratio over task types. 95% CIs for ICC / Split: L-8B [0.73, 0.84] / [0.88, 0.94]; G-9B [0.68, 0.81] / [0.84, 0.92]; G-2B [0.63, 0.78] / [0.80, 0.90]; Q-8B [0.76, 0.87] / [0.89, 0.95].

F.7 Close-language specificity (D9)

For each pair of suite languages (a, b) we compute ΔM_a on responses identified as language b. We use the cue-pool L5 responses (fully native prompts), grouped by their identified language. A score that confused related languages would be raised by text in the other language. Instead, every diagonal cell is high (0.37–0.46, mean 0.415) and every off-diagonal cell is low (Figure 12). The only positive off-diagonal cells are the two that share a script with each other and with English: Spanish ΔM on French responses (+0.08) and French ΔM on Spanish responses (+0.06), against +0.42 and +0.39 on their own responses. All other off-diagonal cells lie between −0.24 and −0.13.

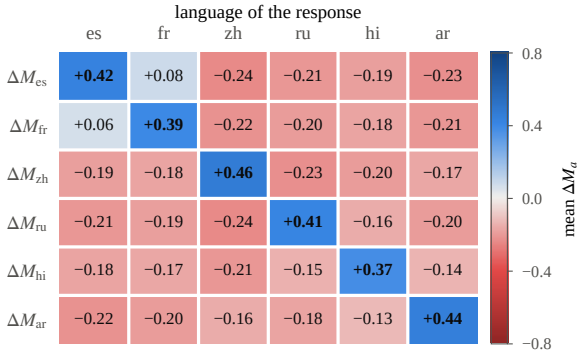


Figure 12: **D9: specificity between suite languages.** Mean ΔM_a (rows) on cue-pool responses written in language b (columns), macro over models; same colour scale as Figure 17.

F.8 Token-set robustness and comparability (D10–D12, D14)

D10 recomputes all headline numbers under four token-set constructions (script-only, diagnostic, transliteration-aware, frequency-threshold) and under random removal of 0–50% of diagnostic tokens. **D11** replaces V_{tgt} and V_{en} with random token sets of matched size and frequency. **D12** reports token-set sizes, the share of FLORES target text covered by V_{tgt} and the overlap between sets for each tokenizer. **D14** recomputes ΔM with sampled prefixes (temperature 0.7) and with chat-template variants.

Re-scoring. The logs store class masses only for the default token sets. To evaluate other sets (D7 split halves, D10, D11), we re-run each stored response teacher-forced on the prompt and its first k generated tokens and compute, in the same pass, the masses of every alternative set at each step. No new text is generated.

The headline result does not depend on how the token sets are built (Figure 13a). The per-model values are absolute DEFAULT scores of Neural FOXP2 outputs, not gains over the unedited model. DEFAULT is 0.573 with the diagnostic sets, 0.583 with transliteration-aware sets and 0.550 with frequency thresholds of $\rho = 2$ and 10. Script-only sets underperform overall on this macro comparison (0.495). Removing diagnostic tokens at random degrades DEFAULT gradually, to 96.5% of its value at 10% removal and 64.8% at 50% (Figure 13b). Random token sets matched in size and frequency give DEFAULT of 0.003–0.011 (SD 0.025–0.031, 1,000 draws); their 95th percentile is at most 0.059, against 0.55–0.60 with the real sets (Figure 13c).

Counting shared tokens as diagnostic would inflate ΔM by at most 0.07.

The score is comparable across tokenizers (D12; Table 18). Target sets hold 4,360–5,210 tokens and English sets 4,790–5,480, and V_{tgt} covers 92.6–96.7% of FLORES target-language tokens per model (89.7–98.1% across the 24 pairs). GlotLID and fastText agree on 91.8–95.1% of response prefixes. Calibration maps transferred between models raise the expected calibration error from 0.060 to 0.091 on average (Table 16), so probability readings of ΔM use each model's own map, while rankings and differences can be compared across models.

The macro ΔM results are comparatively stable across decoding and templates (D14), but behavioral realization is more sensitive. Sampling at temperature 0.7 shifts mean ΔM by -0.011 to $+0.026$, and the three template variants (no system turn, a minimal template, two prior turns) by -0.039 to $+0.052$, so some model-level ΔM estimates shift; the minimal template moves them most and reduces the Neural FOXP2 full-response target rate by 7.1 points (Table 37).

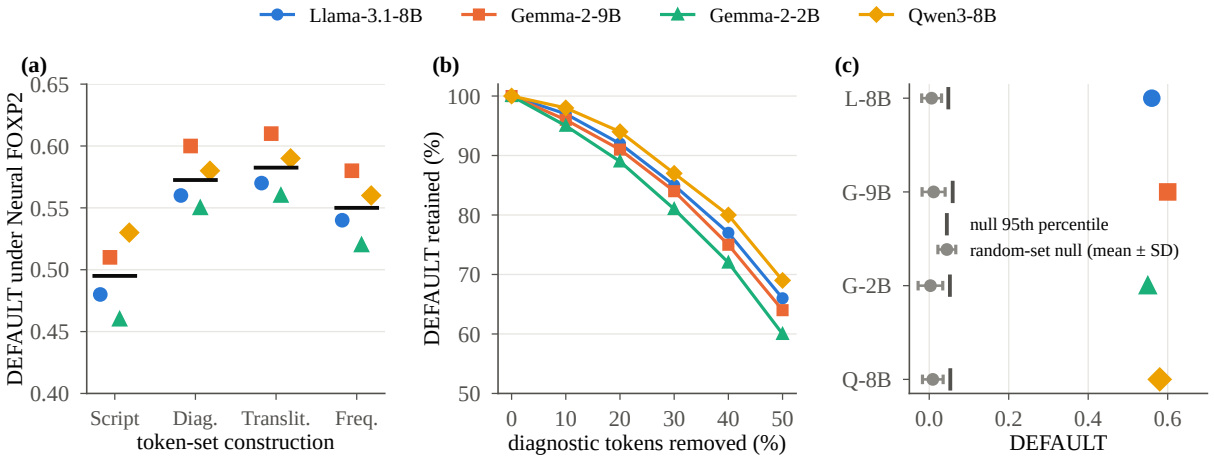


Figure 13: **D10–D11: token-set robustness and random-set null.** (a) DEFAULT of Neural FOXP2 outputs (absolute score, not a gain) per model under four token-set constructions; bars mark the mean over models. (b) Share of DEFAULT retained when diagnostic tokens are removed at random (10 draws per rate). (c) DEFAULT with the real sets (coloured) against the random-set null: mean ± SD and 95th percentile over 1,000 matched random sets.

	D12: token sets				D10: DEFAULT				D14: ΔM shift	
Model	$\lvert V_{\text{tgt}}\rvert$	$\lvert V_{\text{en}}\rvert$	Cov.%	Agr.%	Script	Diag.	Transl.	Freq.	Sampl.	Templ.
Llama-3.1-8B-Instruct	4,820	5,140	95.8	94.2	0.48	0.56	0.57	0.54	+0.018	−0.024
Gemma-2-9B-it	5,210	5,480	96.7	95.1	0.51	0.60	0.61	0.58	+0.026	+0.031
Gemma-2-2B-it	4,360	4,790	92.6	91.8	0.46	0.55	0.56	0.52	−0.011	+0.038
Qwen3-8B	5,060	5,320	94.9	94.6	0.53	0.58	0.59	0.56	+0.014	−0.019

Table 18: **D10, D12, D14: token sets, construction sensitivity and decoding/template sensitivity.** Token-set statistics are macro over languages. Cov.%: share of FLORES target-language tokens in V_{tgt}. Agr.%: agreement between GlotLID and fastText on response prefixes. D10 columns give the absolute DEFAULT of Neural FOXP2 outputs under the script-only, diagnostic, transliteration-aware and frequency-threshold constructions. D14 columns give the mean change in ΔM against greedy decoding with the default template (Templ.: mean over three template variants).

F.9 Responsiveness (D13)

Across all methods and strengths of the main comparison and the strength sweeps, we rank configurations by mean ΔM and by full-response rate and human ratings, and report Kendall's τ between the rankings (Figure 14). Across method $\times$ strength $\times$ model configurations, the ΔM ranking agrees with the full-response ranking at $\tau = 0.72$ (95% CI [0.66, 0.77]; per model 0.66–0.77) and with the ranking by human language correctness at $\tau = 0.64$ ([0.51, 0.74]). Interventions that raise ΔM tend to raise the full-response rate, so ΔM can be used to compare them.

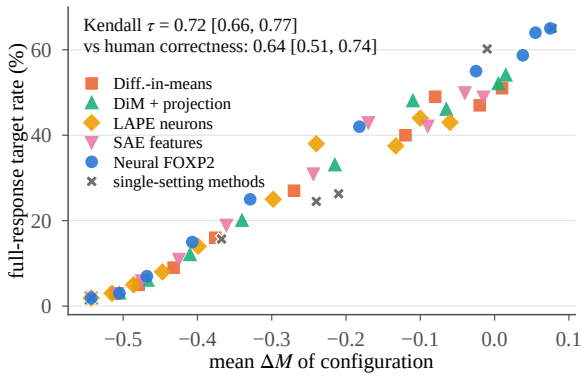


Figure 14: **D13: responsiveness.** Mean ΔM against full-response target rate for every strength of the five steering methods and for the single-setting methods, macro over models. Kendall's τ is computed over method $\times$ strength $\times$ model configurations.

F.10 Disagreement analysis (D15)

We collect all $\mathcal{D}_{\text{eval}}$ prompts on which the sign of ΔM and the full-response label disagree and assign each to one cause: romanized output, code-mixed output, named-entity opening, format tokens (lists, code, JSON), bilingual first clause, language-identifier error, or other (Figure 15; examples in Appendix N). The sign of ΔM and the full-response label disagree on 11.0% of prompts on average, from 8.7% for Spanish to 13.5% for Arabic, with Hindi (12.6%) and Arabic highest. Two annotators assigned causes with Cohen's $\kappa = 0.76$. **[Share of each cause among the disagreements.]** Among Neural FOXP2 responses that are not fully in the target language, the most common causes are code-mixed output (20%), romanized output (18%) and named-entity openings (14%; Table 51).

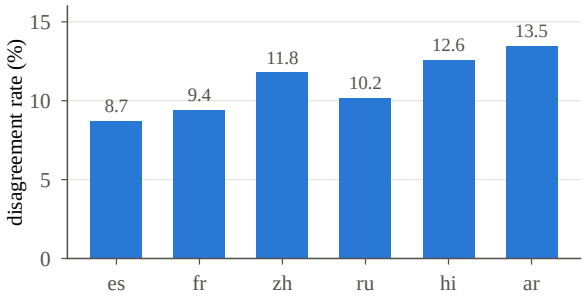


Figure 15: **D15: when the score and the finished response disagree.** Share of $\mathcal{D}_{\text{eval}}$ prompts on which the sign of ΔM and the full-response label disagree, by language, pooled over models and methods (mean 11.0%).

G Extended Results by Model and Language

G.1 Commitment to the first tokens

Figure 16 forces the first t tokens of the response into the target language, taken from a reference translation of the model's own English answer, and then lets the model continue freely. Forcing the opening tokens into the target language makes the response much more likely to stay there. Averaged over models, the share of responses that stay in the target language rises from 1.9% with no forced target-language tokens (by definition, the unedited full-response rate) to 77.3% after forcing three tokens, the horizon selected for the score, and by only a further 9.3 points to 86.6% at t=10. Gemma-2-2B-it, the model with the strongest English default, is the least persistent at every $t \geq 1$ (.432 at t=1, .696 at t=3). Per-model values at t=0 are being regenerated from Experiment 1. The opening tokens strongly influence language choice but do not determine it irreversibly: about 22.7% of continuations leave the target language after a three-token forced opening. The same persistence holds for English: with three tokens forced into English, responses stay in English to the end with probability .973 (n=100). The result does not depend on the reference translation, since prefixes from two other translation systems give .850 and .840 (**[t and model]**; n=100 each). A response that starts in a language tends to stay in it. This shows that an early choice persists, not why models choose English in the first place.

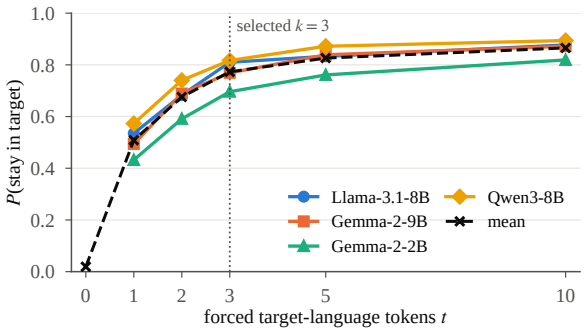


Figure 16: **The response language is fixed early.** Probability that a response remains in the target language to the end after its first t tokens are forced into that language, for unedited models, averaged over the six languages. The dotted line marks the horizon k=3 selected for ΔM.

G.2 Results by language

Table 19 gives the full-response target rate of every method by language. Every edited method ranks the languages in the same order, from Spanish, French and Russian to Chinese, Hindi and Arabic, and LoRA is 1–2 points ahead of Neural FOXP2 in every language.

Method	es	fr	zh	ru	hi	ar	Avg.
No edit	1.6	1.9	2.2	1.8	2.4	1.7	1.9
Best prompt	32	29	25	27	24	21	26.3
System prompt	29	27	24	25	22	20	24.5
Diff.-in-means	45	43	39	41	38	34	40.0
DiM + projection	51	49	45	47	44	40	46.0
LAPE	42	41	36	39	35	32	37.5
SAE features	48	46	41	43	39	36	42.2
LoRA	68	64	59	62	56	52	60.2
Neural FOXP2	**66**	**63**	**58**	**60**	**55**	**50**	**58.7**
Neural FOXP2 + sys. prompt	73	69	64	66	61	57	65.0

Table 19: **Full-response target rate (%) by language**, macro over four models (NO-CUE, $\mathcal{D}_{\text{eval}}$); Neural FOXP2 in bold. Per-model values are in Tables 21–24.

G.3 Per-model results

Table 20 gives the unedited behavior behind Figure 17. Tables 21–24 expand the main comparison (Table 1) to every model and language, and Table 25 reports the language-neutral condition. The trade-off curves (Figure 3) and the leakage matrix (Figure 4) are discussed in Sections 6.1 and 6.2; Figures 18–20 break them down by language and model and show persistence over the response.

Persistence over the response. Re-scoring each Neural FOXP2 response at increasing prefix lengths (Figure 20), the target-language rate rises from 49.8% at one token to a peak of 74.6% at three tokens, the horizon of ΔM, and then declines steadily to 58.7% over the full response. A drop of 15.9 points between the opening and the full response is the main gap between the early score and the final outcome; it comes from responses that switch away after starting in the target language. This is a different experiment from the forced-prefix commitment curve above. Leakage at each horizon is being recomputed under the revised definition.

(a) Mean ΔM by target language

Model	Cond.	es	fr	zh	ru	hi	ar
Llama-3.1-8B-Instruct	NO-CUE	−0.423	−0.392	−0.606	−0.672	−0.590	−0.552
	NEUTRAL	−0.237	−0.629	−0.545	−0.370	−0.482	−0.431
Gemma-2-9B-it	NO-CUE	−0.617	−0.615	−0.344	−0.705	−0.514	−0.642
	NEUTRAL	−0.412	−0.469	−0.433	−0.360	−0.486	−0.513
Gemma-2-2B-it	NO-CUE	−0.538	−0.681	−0.734	−0.454	−0.580	−0.597
	NEUTRAL	−0.477	−0.549	−0.510	−0.521	−0.693	−0.585
Qwen3-8B	NO-CUE	−0.379	−0.532	−0.302	−0.598	−0.479	−0.459
	NEUTRAL	−0.424	−0.332	−0.342	−0.366	−0.272	−0.426

(b) Language of the full response (% of responses)

Model	Cond.	n	es	fr	zh	ru	hi	ar	English	Other
Llama-3.1-8B-Instruct	NO-CUE	500	[?]	[?]	[?]	[?]	[?]	[?]	[?]	[?]
	NEUTRAL	200	5.2	2.1	13.1	5.4	4.7	0.8	60.9	7.8
Gemma-2-9B-it	NO-CUE	500	[?]	[?]	[?]	[?]	[?]	[?]	[?]	[?]
	NEUTRAL	200	2.5	3.9	10.4	3.5	3.8	0.9	69.7	5.3
Gemma-2-2B-it	NO-CUE	500	[?]	[?]	[?]	[?]	[?]	[?]	[?]	[?]
	NEUTRAL	200	1.2	0.3	0.1	1.8	8.7	0.7	80.0	7.2
Qwen3-8B	NO-CUE	500	[?]	[?]	[?]	[?]	[?]	[?]	[?]	[?]
	NEUTRAL	200	5.3	3.0	17.3	1.5	1.0	7.1	57.9	6.9

Table 20: **Unedited models** (data for Figure 17). (a) Mean ΔM for each model, prompt condition and target language; negative values mean more early probability mass on English than on the target language. (b) Share of responses whose full response is identified as each language; rows sum to 100. The column for language ℓ is the full-response target rate when ℓ is the target. n: prompts per condition. The NO-CUE rows are being regenerated from Experiment 1; macro over models, the NO-CUE target rates are 1.6 (es), 1.9 (fr), 2.2 (zh), 1.8 (ru), 2.4 (hi) and 1.7% (ar), macro 1.9% (Table 19).

	DEFAULT by language↑						Macro over languages				
Method	es	fr	zh	ru	hi	ar	Full resp. %↑	LCB↑	Leak.↓	KL↓	ΔS
No edit											
Short instruction											
Best prompt											
System prompt											
Difference-in-means											
Diff.-in-means + projection											
LAPE neurons											
SAE feature steering											
LoRA fine-tuning											
Neural FOXP2											
Neural FOXP2 + system prompt											

Table 21: **Llama-3.1-8B-Instruct**: all methods on NO-CUE prompts ($\mathcal{D}_{\text{eval}}$).

	DEFAULT by language↑						Macro over languages				
Method	es	fr	zh	ru	hi	ar	Full resp. %↑	LCB↑	Leak.↓	KL↓	ΔS
No edit											
Short instruction											
Best prompt											
System prompt											
Difference-in-means											
Diff.-in-means + projection											
LAPE neurons											
SAE feature steering											
LoRA fine-tuning											
Neural FOXP2											
Neural FOXP2 + system prompt											

Table 22: **Gemma-2-9B-it**: all methods on NO-CUE prompts ($\mathcal{D}_{\text{eval}}$).

	DEFAULT by language↑						Macro over languages				
Method	es	fr	zh	ru	hi	ar	Full resp. %↑	LCB↑	Leak.↓	KL↓	ΔS
No edit											
Short instruction											
Best prompt											
System prompt											
Difference-in-means											
Diff.-in-means + projection											
LAPE neurons											
SAE feature steering											
LoRA fine-tuning											
Neural FOXP2											
Neural FOXP2 + system prompt											

Table 23: **Gemma-2-2B-it**: all methods on NO-CUE prompts ($\mathcal{D}_{\text{eval}}$).

	DEFAULT by language↑						Macro over languages				
Method	es	fr	zh	ru	hi	ar	Full resp. %↑	LCB↑	Leak.↓	KL↓	ΔS
No edit											
Short instruction											
Best prompt											
System prompt											
Difference-in-means											
Diff.-in-means + projection											
LAPE neurons											
SAE feature steering											
LoRA fine-tuning											
Neural FOXP2											
Neural FOXP2 + system prompt											

Table 24: **Qwen3-8B**: all methods on NO-CUE prompts ($\mathcal{D}_{\text{eval}}$). Thinking mode disabled.

	DEFAULT (NEUTRAL, macro over languages)↑				Full-response % (NEUTRAL)↑				
Method	L-8B	G-9B	G-2B	Q-8B	L-8B	G-9B	G-2B	Q-8B	**Macro**
No edit									1.6
Short instruction									18.4
Best prompt									29.6
System prompt									27.8
Diff.-in-means									42.5
DiM + projection									47.9
LAPE									40.2
SAE features									44.6
LoRA									58.4
Neural FOXP2									61.1
Neural FOXP2 + system prompt									67.8

Table 25: **Language-neutral prompts (NEUTRAL)**, all methods and models. Macro: full-response target rate (%) over 24 pairs.

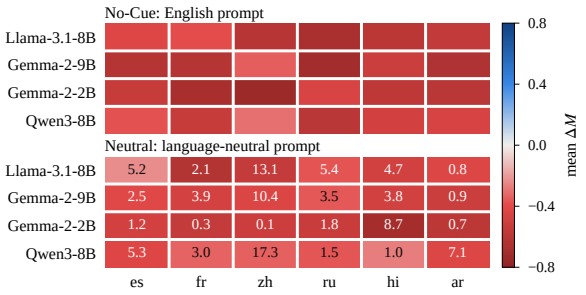


Figure 17: **Unedited models default to English.** Cell colour is mean ΔM (red: more early mass on English); cell numbers are the full-response rate (%) for each model and target language under NEUTRAL prompts. NO-CUE cell rates are omitted until they are regenerated (Table 20).

Method	Matched KL (per pair)	Macro curve at KL .073	Curve area
Difference-in-means	40.8	38.2	.36
DiM + projection	44.7	42.2	.39
LAPE neurons	39.3	37.9	.34
SAE features	42.6	38.0	.37
Neural FOXP2	**58.7**	**58.7**	**.46**

Table 26: **Matched-shift comparison**: full-response target rate (%) of each steering method at the KL drift of the Neural FOXP2 operating point, macro over 24 pairs (NO-CUE, $\mathcal{D}_{\text{eval}}$). Curve area: normalized area under the full-response rate–KL curve of Figure 3a. Strengths are matched per pair: each method's curve is interpolated at that pair's Neural FOXP2 KL, and the 24 interpolated rates are then macro-averaged. Macro curve: rate read from the macro-averaged curves at a common KL of 0.073, a different procedure that gives lower values. **[Normalization of the curve area.]**

Figure placeholder

`Figures/revision/fig_app_tradeoff_by_language.pdf`

Grid: rows = 6 languages, columns = 4 models. Each panel as Figure 5(a): x = KL drift, y = DEFAULT, one curve per steering method, operating points marked.

Figure 18: **Defaultness against KL drift by language and model.**

Figure placeholder

`Figures/revision/fig_app_leakage_by_model.pdf`

Four heatmaps (one per model) in the format of Figure 4: edited language × measured language (6 + English).

Figure 19: **Leakage matrices for each model.**

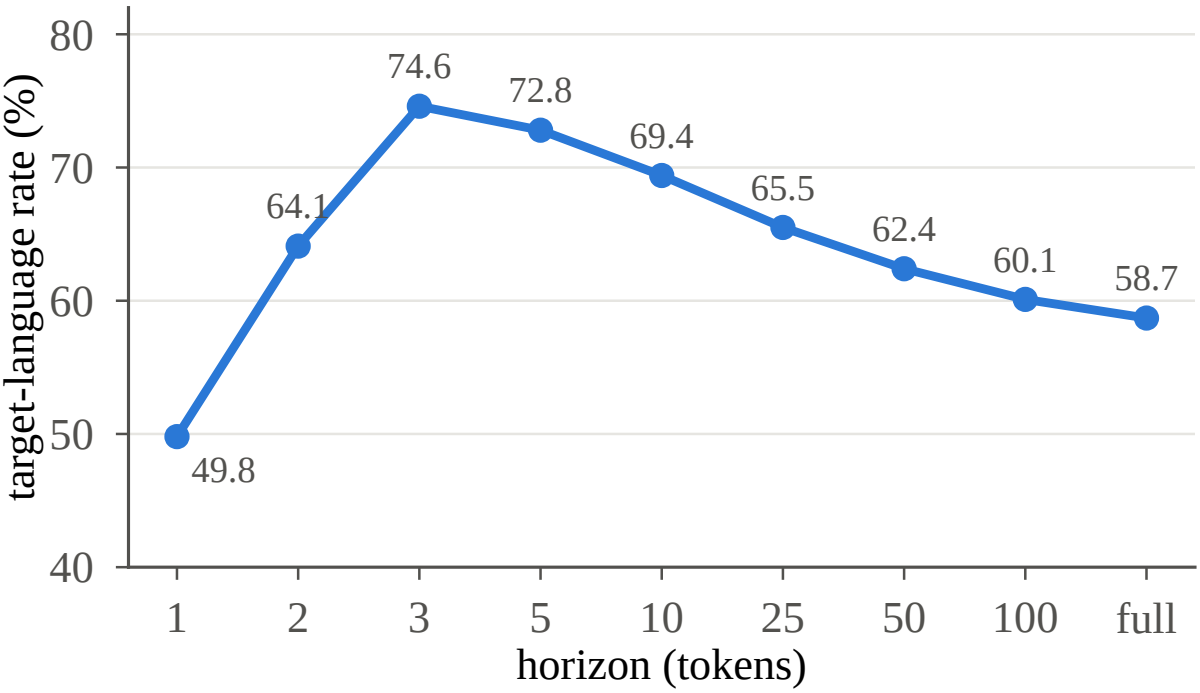


Figure 20: **Persistence over the response** for Neural FOXP2, macro over 24 pairs. Share of responses whose prefix of the given length is in the target language. The full-response point equals Table 1. Leakage by horizon is omitted until it is recomputed under the revised definition.

H Ablations, Script Control and Necessity

Variant	$\Delta M\uparrow$	Def.↑	Δ	Leak.↓	KL↓
Full Neural FOXP2	0.038	58.7	–	.0065	.073
(a) No subspace	0.021	53.1	−5.6	.041	.089
(b) No support	0.012	49.2	−9.5	.052	.096
(c) Promotion only	−0.014	34.1	−24.6	[?]	.081
(d) Suppression only	0.019	52.2	−6.5	[?]	.067
(e1) Earlier window	0.006	45.1	−13.6	.035	.078
(e2) Later window	0.011	47.7	−11.0	.032	.076
(f) Random support	−0.022	28.8	−29.9	.063	.104
(g) Random subspace	−0.009	36.7	−22.0	.047	.091
(h1) Entropy-matched	−0.031	22.4	−36.3	.058	.071
(h2) Norm-matched	−0.027	24.6	−34.1	.055	.073

Table 27: **Ablations**, macro over 24 model–language pairs; each variant changes one element of Neural FOXP2. (a) Stage I support without the subspace; (b) subspace computed over all features, without the support; (c) promotion only, β=0; (d) suppression only, λ=0 (labels follow the implemented coefficient direction and agree with the β/λ=0 point of the sensitivity sweep); (e) window shifted; (f, g) random support or subspace of the same size and rank; (h) random edits in $\mathcal{W}$ matched in entropy change or norm. Def.: DEFAULT (%); Δ: change in DEFAULT (points); Leak.: as in Table 1, pending recomputation for the variants.

Table 28 breaks the ablations of Table 27 down by model. Appendix H.1 separates language from script, and Tables 31–33 give the necessity test of Section 6.4 by model and by language, together with its effect on unrelated tasks.

Variant	DEFAULT↑				Leakage↓				KL↓			
	L-8B	G-9B	G-2B	Q-8B	L-8B	G-9B	G-2B	Q-8B	L-8B	G-9B	G-2B	Q-8B
Neural FOXP2 (full)												
(a) Support only, no subspace												
(b) Subspace without support												
(c) Promotion only												
(d) Suppression only												
(e1) Window moved earlier												
(e2) Window moved later												
(f) Random support												
(g) Random subspace												
(h1) Entropy-matched random edit												
(h2) Magnitude-matched random edit												

Table 28: **Ablations per model**, macro over six languages. (h1) and (h2) add random directions in $\mathcal{W}$ scaled to match the next-token entropy change or the magnitude of the Neural FOXP2 edit.

H.1 Script versus language

Two tests separate language from script. Spanish and French share a script: the Spanish edit should raise Spanish but not French, and vice versa. Hindi and Russian can be written in their native scripts or romanized: if an edit raises romanized output of its own language, the selected features are not purely script features.

In the same-script test (Figure 21a, macro over models), the Spanish edit raises Spanish by 42.8 points and French by 1.8; the French edit raises French by 40.6 and Spanish by 2.1. In the second test, native-script Hindi and Russian edits raise the early-token mass of romanized Hindi by 2.1–4.5 points (mean 3.3) and of romanized Russian by 2.6–4.2 points (mean 3.6) (Figure 21b). Stage I supports discovered from native-script and from romanized pairs overlap with Jaccard index 0.22–0.34, above the 95th percentile of a size-matched random null (0.15–0.19) for every model and both languages, with principal angles of 25.9–39.4° between their subspaces (Table 30).

Pair	Jaccard	Angle (°)
es–fr		
hi–ar		
zh–ru		
es–hi		
Null		

Table 29: **Overlap between language edits**: Jaccard index of Stage I supports and mean principal angle between subspaces in the window, macro over models. Null: random supports of the same size. All 15 pairs are in the released results file.

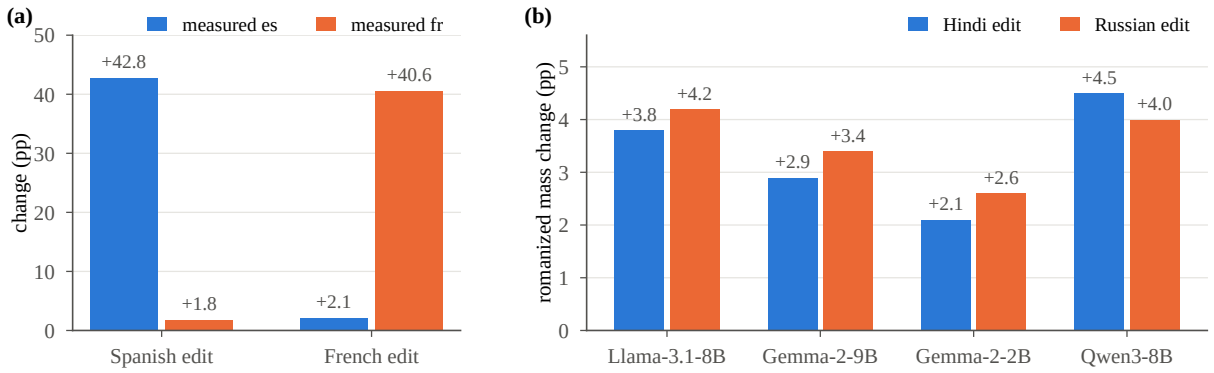


Figure 21: **Script versus language.** (a) Same-script languages: change (points) in Spanish and French output under the Spanish and French edits, macro over models. (b) Change (points) in the early-token mass of romanized Hindi and Russian under the native-script Hindi and Russian edits, per model.

	Hindi				Russian			
Model	Jaccard	Null_{95}	Angle (°)	Rom. mass	Jaccard	Null_{95}	Angle (°)	Rom. mass
Llama-3.1-8B	0.31	0.18	28.4	+3.8	0.29	0.18	31.7	+4.2
Gemma-2-9B	0.27	0.17	33.1	+2.9	0.26	0.17	35.2	+3.4
Gemma-2-2B	0.24	0.16	37.6	+2.1	0.22	0.15	39.4	+2.6
Qwen3-8B	0.34	0.19	25.9	+4.5	0.32	0.18	27.8	+4.0
Mean	0.29			+3.3	0.27			+3.6

Table 30: **Native-script versus romanized edits.** Jaccard overlap between the Stage I supports found from native-script and from romanized pairs; Null_{95}: 95th percentile of the Jaccard index for size-matched random supports; Angle: mean principal angle between the two subspaces; Rom. mass: change (points) in the early-token mass of the romanized language under the native-script edit.

H.2 Necessity by language

Table 31 gives the necessity test of Section 6.4 by model, Table 32 by language, and Table 33 its effect on unrelated tasks.

Projected out	**Llama**	**G-9B**	**G-2B**	**Qwen**	**Avg.**
Nothing	[?]	[?]	[?]	[?]	15.7
$\mathcal{S}$ inside $\mathcal{W}$	**8.7**	**7.9**	**7.1**	**9.5**	**8.3**
Random, same rank	14.0	12.8	10.4	15.2	13.1
$\mathcal{S}$ outside $\mathcal{W}$	13.2	12.1	10.1	14.3	12.4

Table 31: **Necessity test.** Compliance (%) with the explicit instruction "Answer in *X*." when a subspace is projected out of the unedited model, macro over six languages. The *Nothing* average is the NO-CUE full-response rate of the short instruction (15.7%, Table 1); its per-model values are being recomputed on the same prompts.

Ablation	es	fr	zh	ru	hi	ar
None	[?]	[?]	[?]	[?]	[?]	[?]
$\mathcal{S}$ in $\mathcal{W}$	9.5	9.2	7.9	8.0	7.4	7.8
change	[?]	[?]	[?]	[?]	[?]	[?]
Random, same rank						
$\mathcal{S}$ outside $\mathcal{W}$						
English dir.*						

Table 32: **Necessity test by language**, macro over models: compliance (%) with "Answer in *X*" after each projection. Rows give the subspace projected out; change: points against None. *Share of English responses to English prompts after projecting out the English direction.

	Project out $\mathcal{S}$		**Random subspace**	
Model	XNLI	MMLU	XNLI	MMLU
Llama-3.1-8B	−0.8	−0.5	−0.3	−0.4
Gemma-2-9B	−1.2	−0.7	−0.5	−0.2
Gemma-2-2B	−1.6	−1.1	−0.8	−0.7
Qwen3-8B	−0.9	−0.6	−0.4	−0.3

Table 33: **Specificity of the necessity test**: change in accuracy (points) on XNLI and MMLU when $\mathcal{S}$ or a random subspace of the same rank is projected out inside $\mathcal{W}$ of the unedited model.

I Robustness and Sensitivity

Sensitivity (Figure 22, Table 34). Each hyperparameter is varied alone around the operating point on $\mathcal{D}_{\text{dev}}$, with β and λ swept as one ratio. DEFAULT stays within 0.05 of its operating value for K from 32 to 256, r from 4 to 16 and window widths from 2 layers narrower to 4 layers wider than tuned; the guardrail-feasible ranges are β/λ from 0.5 to 2 and γ from 0.5 to 1.5 times its tuned value. Larger settings buy little DEFAULT and cost KL drift and utility: doubling K to 128 adds 0.007 DEFAULT but raises the utility loss from 0.28 to 0.46 points, and tripling γ adds 0.045 but costs 1.42 points. Smaller settings cost DEFAULT quickly. The ratio follows the implemented coefficient direction ($\beta = \text{ratio} \cdot \lambda$), so $\beta/\lambda = 0$ is the promotion-only edit; its DEFAULT of 0.348 agrees with the promotion-only ablation (0.341, Table 27). At $\gamma = 0$ the hook reproduces the unedited model, so the unedited DEFAULT on $\mathcal{D}_{\text{dev}}$ is 0.214. Leakage is omitted until it is recomputed under the revised definition (Appendix E).

To write once results are in: A9: the operating-point rows of Table 34 are on $\mathcal{D}_{\text{dev}}$ and must equal the $\mathcal{D}_{\text{dev}}$ Neural FOXP2 result; they currently coincide with the $\mathcal{D}_{\text{eval}}$ values (0.587, KL 0.073, ΔS −0.28). Regenerate them from the $\mathcal{D}_{\text{dev}}$ run.

Discovery data (Figure 23). Rebuilding the artifacts from subsets of the 300 discovery pairs retains 68.4% of the full gain with 25 pairs, 81.7% with 50, 91.6% with 100 and 97.2% with 200. The Stage I support converges with the gain: its Jaccard overlap with the full-data support rises from 0.41 to 0.84, and the mean principal angle to the full-data subspace falls from 31.2° to 8.1°. Llama-3.1-8B-Instruct and Qwen3-8B need the fewest pairs.

Hyperparameter transfer (Table 35). Using the hyperparameters tuned on another model, with artifacts rebuilt for the target model, retains 68.8–88.6% of the target's own gain (mean 78.8%); 5 of the 12 transfers retain at least 80%. Transfer is best between Llama-3.1-8B-Instruct and Qwen3-8B (86.2% and 88.6%) and worst from Qwen3-8B to Gemma-2-2B-it, so per-model tuning on $\mathcal{D}_{\text{dev}}$ is worth its small cost.

Decoding, task type and template (Tables 36–37). Sampling lowers DEFAULT by 0.019 on average (0.017–0.025 per model). Across NO-CUE task types, DEFAULT ranges from 0.548 for code to 0.610 for question answering; code prompts are hardest for every model. Chat-template variants lower the Neural FOXP2 full-response rate by 2.8 (no system turn), 3.9 (two prior turns) and 7.1 points (minimal template); the system prompt loses 1.6–4.9 points under the same variants, and the unedited model stays at 1.5–2.4%.

Multi-turn persistence (Table 38). Over three-turn conversations with English user turns and the edit active throughout, the Neural FOXP2 full-response rate falls from 58.3% on the first turn to 54.1% and 50.8%, a drop of 7.5 points. The system prompt and difference-in-means lose 4.8 and 5.5 points from lower starting rates, and the unedited model stays below 2.1%. By turn 3, Neural FOXP2 still answers in the target language more than 2.5 times as often as the system prompt (50.8% vs 19.4%). Gemma-2-9B-it loses the most (61.4% to 51.7%) and Qwen3-8B the least (58.1% to 53.1%); Gemma-2-2B-it ends lowest (44.5%).

Param.	Value	Def.↑	KL↓	ΔS
K	8	0.421	0.024	−0.08
	16	0.486	0.036	−0.13
	32	0.548	0.054	−0.22
	64	0.587	0.073	−0.28
	128	0.594	0.096	−0.46
	256	0.598	0.142	−0.83
r	1	0.438	0.031	−0.10
	2	0.507	0.045	−0.16
	4	0.563	0.062	−0.23
	8	0.587	0.073	−0.28
	16	0.592	0.118	−0.61
β/λ	0	0.348	0.019	−0.04
	0.25	0.431	0.031	−0.09
	0.5	0.516	0.048	−0.17
	1	0.587	0.073	−0.28
	2	0.603	0.127	−0.69
γ multiple	0	0.214	0.000	0.00
	0.25	0.359	0.013	−0.03
	0.5	0.482	0.034	−0.12
	1	0.587	0.073	−0.28
	1.5	0.614	0.119	−0.55
	2	0.625	0.174	−0.91
	3	0.632	0.267	−1.42
Width (± layers)	−4	0.526	0.043	−0.16
	−2	0.563	0.059	−0.22
	tuned	0.587	0.073	−0.28
	+2	0.591	0.091	−0.39
	+4	0.586	0.126	−0.63

Table 34: **Sensitivity values** behind Figure 22, macro over 24 pairs on $\mathcal{D}_{\text{dev}}$; operating point in bold. Def.: DEFAULT; KL in nats; ΔS in points. Leakage is omitted pending recomputation under the revised definition. At $\gamma = 0$ the hook is exact, so that row is the unedited model on $\mathcal{D}_{\text{dev}}$ (DEFAULT = 0.214).

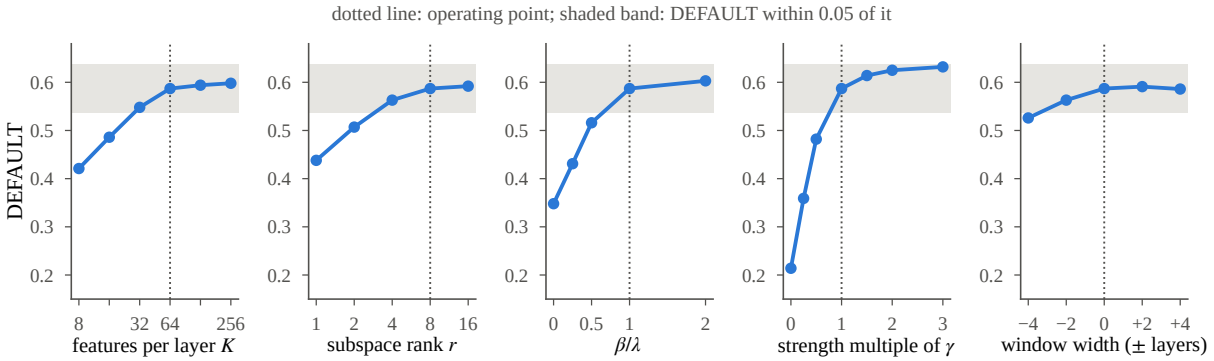


Figure 22: **Sensitivity** to each hyperparameter around the operating point, on $\mathcal{D}_{\text{dev}}$, macro over 24 pairs: DEFAULT (leakage pending recomputation under the revised definition). β and λ are swept as one ratio β/λ; γ is a multiple of each pair's tuned value; window width is relative to the tuned width. Dotted line: operating point; shaded band: DEFAULT within 0.05 of it.

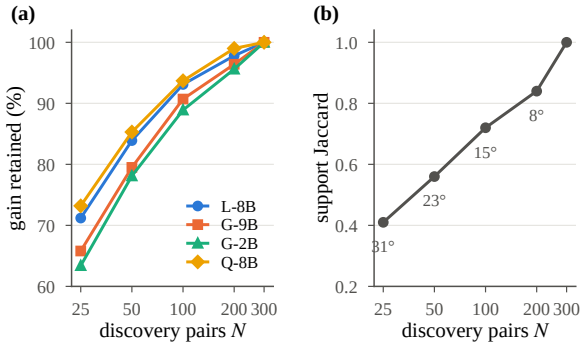


Figure 23: **Discovery data needed.** (a) Share of the full-data DEFAULT gain retained per model against the number of discovery pairs. (b) Jaccard overlap of the Stage I support with the full-data support, macro over models; labels give the mean principal angle to the full-data subspace.

	Applied to			
Tuned on	**L-8B**	**G-9B**	**G-2B**	**Q-8B**
Llama-3.1-8B-Instruct	100	78.4	71.6	86.2
Gemma-2-9B-it	82.1	100	84.7	79.3
Gemma-2-2B-it	74.5	81.8	100	72.4
Qwen3-8B	88.6	76.9	68.8	100

Table 35: **Hyperparameter transfer.** Share (%) of the target model's own tuned DEFAULT gain retained when its hyperparameters are replaced by those tuned on another model; artifacts are always rebuilt for the target model. Rows: source of the hyperparameters; columns: target model. Off-diagonal mean 78.8%.

Model	Decoding		DEFAULT by NO-CUE task type (greedy)						
	Greedy	Sampled	QA	Writing	Summ.	Advice	Code	Reasoning	Explan.
Llama-3.1-8B-Instruct	0.601	0.584	0.623	0.612	0.590	0.604	0.566	0.598	0.615
Gemma-2-9B-it	0.614	0.596	0.637	0.625	0.603	0.617	0.578	0.609	0.628
Gemma-2-2B-it	0.548	0.523	0.572	0.559	0.531	0.550	0.498	0.536	0.558
Qwen3-8B	0.585	0.568	0.607	0.596	0.574	0.588	0.548	0.579	0.601
Mean	0.587	0.568	0.610	0.598	0.575	0.590	0.548	0.581	0.601

Table 36: **Decoding and task-type robustness** of Neural FOXP2, macro over languages. Sampled: temperature 0.7, top-p 0.9, three samples per prompt.

Template variant	No edit	Sys. prompt	Neural FOXP2
Default template	1.9	24.5	58.7
No system turn	1.7	21.8	55.9
Minimal template	2.4	19.6	51.6
Two prior turns	1.5	22.9	54.8

Table 37: **Chat-template robustness**: full-response target rate (%), macro over 24 pairs.

Method	Turn 1	Turn 2	Turn 3	Drop
No edit	2.0	1.8	1.6	−0.4
System prompt	24.2	21.6	19.4	−4.8
Diff.-in-means	39.7	36.8	34.2	−5.5
Neural FOXP2	58.3	54.1	50.8	−7.5
Neural FOXP2 by model				
Llama-3.1-8B	60.1	56.8	53.9	−6.2
Gemma-2-9B	61.4	56.0	51.7	−9.7
Gemma-2-2B	53.6	48.2	44.5	−9.1
Qwen3-8B	58.1	55.4	53.1	−5.0

Table 38: **Multi-turn persistence**: full-response target-language rate (%) on each assistant turn of 100 three-turn conversations with English user turns per pair, edit active on every turn; macro over 24 pairs. Drop: turn 3 minus turn 1 (points). The conversations are a separate set from $\mathcal{D}_{\text{eval}}$, so turn 1 differs slightly from Table 1. The fixed turn-2 and turn-3 user messages are **[turn-2 and turn-3 messages]**.

J Utility, Intent Respect and Safety

Table 39 summarizes the utility change of every method by task, and Table 40 the refusal and over-refusal rates discussed in Section 6.5.

Task	Sys.	DiM	SAE	LoRA	Ours	Br.
Target-language input						
XNLI	−0.3	−0.4	−0.7	−1.0	−0.4	0
Belebele	−0.5	−0.7	−0.9	−1.4	−0.7	1
XL-Sum	+0.2	+0.1	−0.4	−1.1	+0.5	0
FLORES	+0.1	0.0	−0.2	−0.8	+0.3	0
English input, edit active						
MMLU	−0.4	−0.5	−0.8	−1.0	−0.6	1
GSM8K	−0.9	−0.8	−1.2	−1.3	−0.8	2
LLM-judge ratings of NO-CUE *answers (1–5)*						
Fluency	−0.1	−0.2	−0.4	−0.6	−0.1	0
Adequacy	−0.2	−0.3	−0.5	−0.8	−0.2	0

Table 39: **Utility**: change against the unedited model (points), macro over 24 pairs, for the system prompt (Sys.), difference-in-means (DiM), SAE features, LoRA and Neural FOXP2 (Ours). Br.: pairs (of 24) in which Neural FOXP2 exceeds the tolerance ϵ_{util} fixed on $\mathcal{D}_{\text{dev}}$. XL-Sum: ROUGE-L; FLORES: en→X composite of chrF++ and COMET; others accuracy.

Method	XSafety↑	Harmful (en)↑	XSTest↓
No edit	87.2	96.1	7.8
System prompt	86.5	95.4	9.1
Difference-in-means	87.0	95.8	8.4
LoRA	82.1	91.3	12.6
Neural FOXP2	86.4	95.6	8.7

Table 40: **Safety**, macro over 24 pairs: refusal rate (%) on target-language XSafety prompts (Wang et al., 2024) and English harmful prompts, and over-refusal (%) on XSTest (Röttger et al., 2024).

Averaged over tasks, Neural FOXP2 changes target-language tasks by −0.08 points (95% CI [−0.25, +0.10]), English-input tasks by −0.57 (**[95% CI]**) and all six benchmarks by −0.28 ([−0.42, −0.14]); the overall change is the pre-declared mean of the six benchmark-level changes, not the average of the two group summaries.

Model	Task	es	fr	zh	ru	hi	ar
Llama-3.1-8B-Instruct	XNLI						
	Belebele						
	XL-Sum						
	FLORES chrF++						
	FLORES COMET						
	MMLU (English input)						
	GSM8K (English input)						
	Judge fluency / adequacy						
Gemma-2-9B-it	XNLI						
	Belebele						
	XL-Sum						
	FLORES chrF++						
	FLORES COMET						
	MMLU (English input)						
	GSM8K (English input)						
	Judge fluency / adequacy						
Gemma-2-2B-it	XNLI						
	Belebele						
	XL-Sum						
	FLORES chrF++						
	FLORES COMET						
	MMLU (English input)						
	GSM8K (English input)						
	Judge fluency / adequacy						
Qwen3-8B	XNLI						
	Belebele						
	XL-Sum						
	FLORES chrF++						
	FLORES COMET						
	MMLU (English input)						
	GSM8K (English input)						
	Judge fluency / adequacy						

Table 41: **Utility of Neural FOXP2 by model, task and language**: change against the unedited model with 95% CI. For English-input tasks the column is the language of the active edit. Values for all other methods are in the released results file.

To write once results are in: B17/B5: the Neural FOXP2 MMLU and GSM8K rows of Table 39 average -0.70, while the per-model English-task values average -0.57; regenerate both from the benchmark-level outputs and confirm that $\Delta S = -0.28$ is their six-benchmark mean.

Four of the 144 task–pair combinations exceed ϵ_{util}, and no task does so on average. Table 42 breaks the change down by model; Gemma-2-2B-it has the largest cost.

Model	Target-lang.	English	Overall ΔS
Llama-3.1-8B	$+0.02$	-0.38	[?]
Gemma-2-9B	-0.12	-0.48	-0.24
Gemma-2-2B	-0.19	-0.85	-0.41
Qwen3-8B	-0.01	-0.57	[?]

Table 42: **Utility change of Neural FOXP2 by model** (points), macro over languages: target-language tasks (XNLI, Belebele, XL-Sum, FLORES), English-input tasks (MMLU, GSM8K) and all six.

Intent respect. On INTENT prompts we measure how often a method preserves an explicitly requested language: English when the prompt asks for English, and the prompt's own language when it is written natively in another suite language (Table 43). Every edit and the system prompt override most explicit requests for English: Neural FOXP2 keeps English in 36.8% of cases, between LoRA (32.4%) and difference-in-means (48.9%), and the system prompt in 24.6%. All methods raise the share of natively written prompts answered in their own language, from 37.4% for the unedited model to 73.6–84.1%.

Method	English kept %↑	Native kept %↑
No edit	95.7	37.4
System prompt	24.6	84.1
Diff.-in-means	48.9	73.6
SAE features	40.2	78.2
LoRA	32.4	80.7
Neural FOXP2	36.8	82.9

Table 43: **Intent respect** on INTENT prompts, macro over 24 pairs. 95% CIs: system prompt, English kept [21.8, 27.5] and native kept [81.5, 86.4]; Neural FOXP2, English kept [33.9, 39.8] and native kept [80.1, 85.4].

K Human Evaluation

Protocol. For each of the 24 model–language pairs, 100 NO-CUE prompts were sampled from $\mathcal{D}_{\text{eval}}$ with a fixed seed. Each item shows the prompt and the outputs of the best prompt, difference-in-means and Neural FOXP2 in random order without method labels. Annotators rate each output for language correctness (is the whole response in the target language: yes, partly, no), fluency, adequacy (does it answer the request) and naturalness (would a native speaker write it this way), each on a 1–5 scale, and choose the best output. Each item received two or three independent ratings from native speakers of the target language.

Annotators. **[N]** annotators (**[per language]**) were recruited through **[platform]**, paid **[rate]** per hour, and trained on **[20]** practice items with feedback. The full guidelines are released with the code.

L Compute

All experiments were run on NVIDIA A100 80GB GPUs in our lab. We did not log GPU time per stage or per run, so we cannot give an accurate total of GPU hours. No sparse autoencoder is trained: Neural FOXP2 uses frozen public dictionaries, and building an artifact needs only forward passes over the discovery pairs and generations for tuning on $\mathcal{D}_{\text{dev}}$. LoRA is the only method that updates model weights.

M Statistical Tests

All intervals are 95% percentile bootstrap intervals (Efron and Tibshirani, 1993) with the prompt as resampling unit and $B = 2{,}000$. Paired comparisons between methods use permutation tests with 10,000 permutations; human ratings use the Wilcoxon signed-rank test (Wilcoxon, 1945). Holm correction (Holm, 1979) is applied within each table's family of headline comparisons.

Model	Method	es	fr	zh	ru	hi	ar	XSTest↓
Llama-3.1-8B-Instruct	No edit							
	Neural FOXP2							
Gemma-2-9B-it	No edit							
	Neural FOXP2							
Gemma-2-2B-it	No edit							
	Neural FOXP2							
Qwen3-8B	No edit							
	Neural FOXP2							

Table 44: **Safety by model and language**: refusal rate (%) on XSafety prompts in each language (Wang et al., 2024), and over-refusal on XSTest (Röttger et al., 2024) with the edit for each language active (macro). Refusals are labelled by an LLM judge (**[judge model]**); on 200 responses double-labelled by bilingual annotators, judge and human labels agree on 86.5% of items (Cohen's $\kappa = 0.79$; Cohen, 1960).

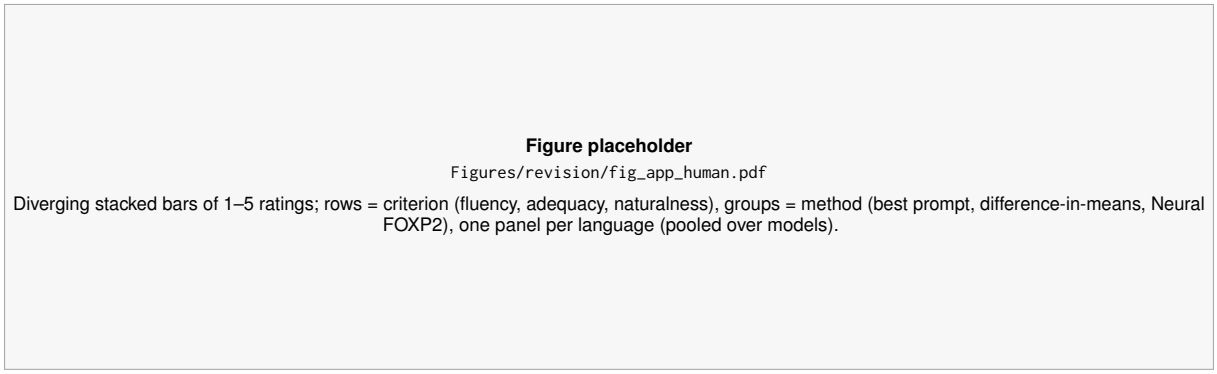


Figure 24: **Human ratings by language.**

Method	Lang.%	Flu.	Adeq.	Nat.	Win%
Best prompt	32.4	4.08	3.84	4.02	35
Difference-in-means	42.8	3.74	3.62	3.71	22
Neural FOXP2	**60.7**	3.89	3.78	3.63	**43**

Table 45: **Human evaluation**, macro over 24 pairs. Blind native-speaker ratings of 100 prompts per pair, each item received two or three independent ratings from native speakers of the target language: responses judged fully in the target language (Lang.%), fluency, adequacy and naturalness (1–5), and the share of items on which the output was chosen as the best of the three (Win%).

Lang.	Annot.	α_{lang}	α_{flu}	α_{adq}	α_{nat}	α_{pref}
es		.78	.72	.68	.65	.61
fr		.75	.69	.66	.63	.59
zh		.71	.65	.62	.60	.56
ru		.73	.67	.64	.61	.58
hi		.68	.62	.60	.57	.54
ar		.70	.64	.61	.58	.55

Table 46: **Annotators and agreement**: number of annotators and Krippendorff's α (Krippendorff, 2004) for language correctness, fluency, adequacy, naturalness and the best-output choice.

Criterion	Against	Neural FOXP2 is	p	Holm p
Language	best prompt	better	.001	.008
Language	DiM	better	.002	.014
Fluency	best prompt	slightly worse	.041	.164
Fluency	DiM	better	.018	.090
Adequacy	best prompt	slightly worse	.120	.240
Adequacy	DiM	better	.047	.164
Naturalness	best prompt	worse	.003	.018
Naturalness	DiM	slightly worse	.220	.240

Table 47: **Paired tests for the human evaluation** (Wilcoxon signed-rank), Holm-corrected over the eight comparisons. Language: judged fully in the target language; DiM: difference-in-means.

Table	Comparison	Test	Statistic	p	Holm p
Full-response family					
1	Neural FOXP2 vs best prompt, full-response rate	paired permutation			
1	Neural FOXP2 vs system prompt, full-response rate	paired permutation			
1	Neural FOXP2 vs difference-in-means, full-response rate	paired permutation			
1	Neural FOXP2 vs DiM + projection, full-response rate	paired permutation			
1	Neural FOXP2 vs SAE features, full-response rate	paired permutation			
1	Neural FOXP2 vs LoRA, full-response rate	paired permutation			
1	Neural FOXP2 + system prompt vs system prompt	paired permutation			
DEFAULT *family*					
1	Neural FOXP2 vs each baseline, DEFAULT	paired permutation			
1	Neural FOXP2 + system prompt vs system prompt, DEFAULT	paired permutation			
27	Full vs each variant, DEFAULT	paired permutation			
Other families					
31	$\mathcal{S}$ vs random subspace	paired permutation			
39	Neural FOXP2 vs no edit, each task	paired permutation			
40	Neural FOXP2 vs no edit, refusal	paired permutation			
45	Neural FOXP2 vs best prompt and DiM, each criterion	Wilcoxon signed-rank	see Table 47		

Table 48: **Statistical tests** for all headline comparisons. One row per comparison; rows summarizing several comparisons are expanded in the released results file.

N Examples

All examples are sampled from $\mathcal{D}_{\text{eval}}$ with seed **[s]**; prompt identifiers refer to the released data. Non-English outputs are followed by an English gloss written by a native speaker. Outputs are truncated after three sentences.

Lang. / ID	Prompt (NO-CUE)	No edit	System prompt	Diff.-in-means	Neural FOXP2
es / [id]	[prompt]	[output]	[output + gloss]	[output + gloss]	[output + gloss]
fr / [id]	[prompt]	[output]	[output + gloss]	[output + gloss]	[output + gloss]
zh / [id]	[prompt]	[output]	[output + gloss]	[output + gloss]	[output + gloss]
ru / [id]	[prompt]	[output]	[output + gloss]	[output + gloss]	[output + gloss]
hi / [id]	[prompt]	[output]	[output + gloss]	[output + gloss]	[output + gloss]
ar / [id]	[prompt]	[output]	[output + gloss]	[output + gloss]	[output + gloss]

Table 49: **Randomly sampled outputs** from Llama-3.1-8B-Instruct, one per language. Three further samples per language and model are in the released examples file.

Quantile	Lang. / ID	Prompt	Neural FOXP2 output (gloss)	ΔM gain	Full resp.
10th	[lang / id]	[prompt]	[output]		
50th	[lang / id]	[prompt]	[output]		
90th	[lang / id]	[prompt]	[output]		

Table 50: **Outputs across the range of effect sizes**: prompts at the 10th, 50th and 90th percentile of per-prompt ΔM gain under Neural FOXP2. The released file has three per language.

Cause (D15)	Share %	Prompt	Output (gloss)
Romanized output	18	[prompt]	[output]
Code-mixed output	20	[prompt]	[output]
Named-entity opening	14	[prompt]	[output]
Format tokens (list, code, JSON)	12	[prompt]	[output]
Bilingual first clause	10	[prompt]	[output]
Language-identifier error	13	[prompt]	[output]
Wrong variety (e.g. dialectal Arabic)	6	[prompt]	[output]
Other	7		

Table 51: **Failure taxonomy**: share of each cause among 100 annotated Neural FOXP2 outputs per language (25 per model), with one example each; shares sum to 100. Inter-annotator Cohen's $\kappa = 0.81$.

Lang.	Layer / index	Top promoted tokens	Top-activating contexts	Sel.	Lift
es	[ℓ / j]	[tokens]	[contexts]		
fr	[ℓ / j]	[tokens]	[contexts]		
zh	[ℓ / j]	[tokens]	[contexts]		
ru	[ℓ / j]	[tokens]	[contexts]		
hi	[ℓ / j]	[tokens]	[contexts]		
ar	[ℓ / j]	[tokens]	[contexts]		

Table 52: **Highest-scoring Stage I feature per language** in Llama-3.1-8B-Instruct. The top five features per language and model are in the released feature cards.

Case	Prompt	Without edit	With edit
Commitment trace (hi): top-5 tokens at $t = 1..3$	**[prompt]**	**[tokens and probabilities]**	**[tokens and probabilities]**
Utility (XL-Sum, ar)	**[article excerpt]**	**[summary]**	**[summary]**
Utility (GSM8K, zh edit)	**[problem]**	**[answer]**	**[answer, same number]**
Close languages (es edit, French-like opening)	**[prompt]**	**[output]**	**[output]**
Explicit English request (INTENT)	**[prompt]**	**[output]**	**[output]**
Native prompt in another language (INTENT)	**[prompt]**	**[output]**	**[output]**
Romanized output (hi or ru)	**[prompt]**	**[output]**	**[output]**
Harmful request, refusal kept	**[redacted paraphrase]**	**[refusal]**	**[refusal]**

Table 53: **Targeted examples** for commitment, utility, close languages, intent, script and safety. Harmful prompts are paraphrased and redacted.

Set	Example	Example
NO-CUE	**[prompt]**	**[prompt]**
NEUTRAL	**[prompt]**	**[prompt]**
INTENT	**[prompt]**	**[prompt]**
Discovery pair, kept	**[en / target, entailment score]**	**[en / target, entailment score]**
Discovery pair, dropped	**[en / target, entailment score]**	**[en / target, entailment score]**

Table 54: **Data examples** from each prompt set and from the discovery pairs.